CHEMISTRY
CLASS 10

Five papers. Complete worked solutions.

FBISE SSC-II • Exam preparation • 2026

All short and long questions, including every OR part. Balanced equations, worked calculations, labelled diagrams, structural formulae and clear notes on errors in the printed papers.

01 Model Paper — Chemistry SSC-II

02 1st Annual 2026 — Paper B

03 1st Annual 2026 — Paper D

04 1st Annual 2026 — Paper M

05 2nd Annual 2026 — Chemistry SSC-II

Made by Fakhar STEM Sphere

An independent preparation resource. Answers are written in exam-focused English; this is not an official FBISE marking scheme.

How to use this book

Each paper begins with its original question pages, followed by complete solutions in the same question order. Both alternatives are included wherever the paper gives an OR choice.

For numerical answers, write the balanced equation, formula, substitution, working and final answer with units. For reactions, include conditions when relevant. For diagrams, label electrodes, polarity, solutions and electron or ion movement.

Correction notes are part of the preparation material. Do not memorise a forced result for a question whose data are inconsistent.

Units used: dm^3 = litre; mol dm^{-3} = molarity; at RTP, molar gas volume is taken as 24 dm^3 mol^{-1}. Atomic masses follow the papers, except where a printed value is explicitly corrected.

Contents and navigation: use the PDF bookmarks to jump to any of the five papers. Printed page numbers appear in the footer.

Scope: the supplied documents contain Sections B and C. Section A MCQ papers were not supplied.

Editorial approach: all substantive details from the worked answers in the conversation are retained, with equations redrawn clearly and repeated answers written out in each paper. Relevant scientific corrections are preserved rather than silently changing the source question.

PAPER 01
Model Paper — Chemistry SSC-II

Original question paper

The supplied question pages follow. Complete worked solutions begin immediately afterwards.

Federal Board SSC-II Examination
Chemistry Model Question Paper
(Curriculum 2022-2023) Scheme of Studies 2006

Time allowed: 2:40Hours Total Marks Section B & C: 53

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the **E-sheet**. Write your answers on the allotted /given spaces.

SECTION – B (Marks 33)

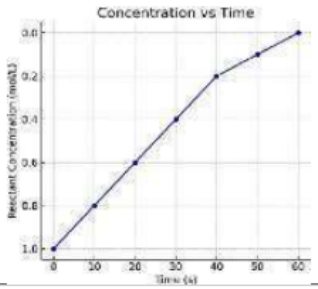
Q. 2	Attempt the following questions (11x3 = 33)				
(i)	Based on the given standard electrode potential (E°) values, arrange the following metals in order of increasing reactivity. Explain how electrode potential relates to metal reactivity. $Mg^{2+} + 2e^- \rightarrow Mg$ E°= -2.37 $Cu^{2+} + 2e^- \rightarrow Cu$ E°= +0.34 $Ni^{2+} + 2e^- \rightarrow Ni$ E°= -0.25	1+2	OR	A gas sample has a volume of 72 dm^3 at room temperature and pressure. How many moles of gas does it contain?	03
(ii)	The Contact Process is an industrial method for sulfuric acid production. a. Write the balanced symbol equation for conversion of sulfur dioxide to sulfur trioxide in the Contact process. b. Identify the sources of sulfur dioxide and oxygen in this process. c. What are the typical conditions for the conversion of sulfur dioxide to sulfur trioxide?	1+1 +1	OR	a. What is molarity? How can you convert molarity to g/dm^3? b. What is the molarity of a solution made by dissolving 98 g of H_2SO_4 in 1 dm^3 of water? (Molar mass of H_2SO_4 = 98 g/mol)	1.5+ 1.5
(iii)	A sample of calcium carbonate ($CaCO_3$) contains 20 grams of calcium. What is the mass of carbon dioxide (CO_2) produced when the calcium carbonate is heated?	03	OR	Draw structural formula of following a. Pent-2-ene b. But-1-ol c. Pentanoic acid	1+1+ 1
(iv)	a. What is a polymer? b. Name the basic building blocks of polymers. c. Which polymer is commonly used to make plastic bottles?	1+1 +1	OR	Draw the structure showing how protein molecule is formed, clearly highlighting the peptide linkages within the structure.	2+1
(v)	Write the chemical equations for the reactions of: a. Zinc with steam. b. Magnesium with hydrochloric acid. c. Sodium with cold water.	1+1 +1	OR	a) What are the reactants in a hydrogen-oxygen fuel cell? b) Which electrode in a hydrogen-oxygen fuel cell undergoes oxidation c) Write the overall reaction equation for a hydrogen-oxygen fuel cell.	03
(vi)	a. What is esterification? Provide a general reaction equation to illustrate the process. b. Why is esterification considered a condensation reaction?	2+1	OR	How does the lattice structure of salts contribute to their physical properties?	03
(vii)	Complete the following reactions: a. CH_3-CH_2-Br + alcoholic KOH → Y b. Y + Br_2 → Z	1.5 + 1.5	OR	Why are SO_2 and CO_2 classified as acidic oxides and CuO and CaO considered basic oxides?	1.5+ 1.5

(viii)	a. Why is petroleum considered a mixture? Give three examples of a hydrocarbon found in petroleum.	1.5 + 1.5	OR	Ethene and ethane are hydrocarbons which belong to different classes. How can they be distinguished? Write at least two tests in each case for the support of your answer.	1+2
(ix)	Give reactions to illustrate how unburned hydrocarbons contribute to the formation of peroxyacetyl nitrates (PANs) ? .	1+1 +1	OR	a. What are the primary fossil fuels used for energy production? b. Identify the fossil fuel that is mainly composed of methane. Among coal, petroleum, and natural gas, which is considered the cleanest-burning fuel	1+1 +1
(x)	Vitamins are important for our health. Summarize their sources and importance in health.	1.5 + 1.5	OR	Explain the combustion process of alcohols and discuss why they are considered cleaner fuels compared to fossil fuels?	1.5 + 1.5
(xi)	Zinc plating is carried out to prevent iron from corrosion. Describe its complete process and chemical reactions taking place at anode and cathode.	1.5 + 1.5	OR	How will you explain the formation of ethanol by: a. Fermentation and catalytic addition of steam. Explain your answer with balanced chemical equations.	1.5+ 1.5

Section – C (Marks 20)

Note: Attempt all questions. Marks for each question are given. (4 x 05=20)

Q.3	Identify the functional groups in the following compounds: a. $HCOOCH_3$ b. CH_3-C ≡ C-CH_3 c. C_3H_6 d. $C_4H_{11}OH$	1×4	O R	Describe the general chemical properties of metals with reference to their reactivity with: i. **Dilute acids** ii. **Cold water** iii. **Steam** iv. **Oxygen** Provide **balanced chemical equations** to illustrate each type of reaction.	1× 4
Q.4	Carbon dioxide (CO_2) is produced by the reaction of carbon monoxide (CO) with oxygen gas (O_2) according to the balanced equation: $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$ If **28 g of carbon monoxide (CO)** reacts with an excess of oxygen gas, answer the following: a. Calculate the number of moles of carbon monoxide (CO) used. b. Determine the number of moles of carbon dioxide (CO_2) produced. c. Find the mass of carbon dioxide (CO_2) formed. Calculate the volume of carbon dioxide (CO_2) produced at room temperature and pressure (RTP), where 1 mole of gas = 24 dm^3.	1.5 ×4		A student is working in a lab and has been assigned a task to examine an unknown hydrocarbon. The compound has been isolated from a crude oil sample and has a molar mass of 70 g/mol. He obtained the following data: - Mass of carbon: 10.0 g - Mass of hydrogen: 3.0 g - Mass of nitrogen: 2.0 g Calculate its empirical and molecular formula.	4+ 2

Q.5	a. What does the rate of reaction mean, and how can it be measured? b. How can a graph of "time vs. reactant concentration" be used to determine the rate of reaction? c. Interpret the given graph and explain how the reaction rate changes over time. 	2+2 +2	O R	Metals X, Y, and Z were tested for their reactivity with dilute acid and water. The results are given below: ➢ Metal X reacts with cold water, forming hydrogen gas. ➢ Metal Y does not react with water but reacts with hydrochloric acid, producing bubbles of hydrogen gas. Metal Z does not react with water or acid. A arrange these metals in order of reactivity and explain how the observations support your answer.	3+ 3
Q.6	a) Draw a schematic diagram for the electrolysis of dilute copper(II) sulphate using inert or copper electrodes. b) Label the anode, cathode, electrolyte, and the direction of ion movement. Describe the observations and products formed at each electrode	1+1 +2	O R	A polymer is formed by the reaction between a dicarboxylic acid and a diamine. a) What type of polymerization occurs in this reaction? Explain. b) Determine the structure of the repeating unit in the resulting polymer. c) Identify the functional group that forms during the polymerization process. Name a well-known synthetic polyamide and mention one of its applications.	04

Model Paper — Chemistry SSC-II

Complete solutions • Sections B and C

SECTION B — Q2: Short questions (33 marks)

Q2(i) Arrange Mg, Cu and Ni in increasing reactivity; explain electrode potential.

Cu < Ni < Mg

Given standard reduction potentials: $Mg^{2+}/Mg = -2.37$ V; $Ni^{2+}/Ni = -0.25$ V; $Cu^{2+}/Cu = +0.34$ V. A more negative reduction potential indicates that the metal loses electrons more readily, so its reactivity is greater. Magnesium is most reactive; nickel is intermediate; copper is least reactive.

OR — How many moles are in 72 dm³ of gas at RTP?

$n = V \div V_m = 72 \div 24 = 3.0$ mol

Answer: 3.0 mol. At RTP, one mole occupies 24 dm³.

Q2(ii) Contact Process: equation, gas sources and conditions.

(a) Conversion of sulfur dioxide to sulfur trioxide:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(b) Sources: SO_2 is obtained by burning sulfur in air; oxygen is supplied by air.

$S(s) + O_2(g) \rightarrow SO_2(g)$

(c) Conditions: approximately 450°C, 1–2 atm, and vanadium(V) oxide (V_2O_5) catalyst.

OR — Define molarity; convert it to g/dm³; calculate molarity of 98 g H_2SO_4 in 1 dm³.

Molarity is the number of moles of solute in **one cubic decimetre of solution**. Unit: mol dm^{-3}.

M = moles of solute ÷ solution volume in dm³

Concentration in g dm^{-3} = molarity × molar mass

$n(H_2SO_4) = 98 \div 98 = 1.0$ mol

$M = 1.0 \div 1 = 1.0$ mol dm^{-3}

The question says “1 dm³ of water”. The expected answer assumes 1 dm³ of final solution. Exact molarity requires the final solution volume.

Q2(iii) $CaCO_3$ contains 20 g calcium. Find CO_2 produced on heating.

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ [heat]

Each mole of calcium carbonate contains one mole of calcium and yields one mole of CO_2.

$n(Ca) = 20 \div 40 = 0.50$ mol

$n(CO_2) = 0.50$ mol

$m(CO_2) = nM_r = 0.50 \times 44 = 22$ g

Answer: 22 g CO_2.

OR — Structural formulae of pent-2-ene, but-1-ol and pentanoic acid.

Compound	Condensed structural formula
Pent-2-ene	$CH_3–CH=CH–CH_2–CH_3$
Butan-1-ol	$CH_3–CH_2–CH_2–CH_2–OH$
Pentanoic acid	$CH_3–CH_2–CH_2–CH_2–C(=O)–OH$

Q2(iv) Define polymer; name building blocks; identify bottle-making polymer.

(a) A polymer is a large molecule formed by joining many small repeating units.

(b) Its small molecular building blocks are **monomers**.

(c) Polyethylene terephthalate (PET) is commonly used for drinks bottles.

OR — Show protein formation and peptide linkages.

Amino acids join through **condensation reactions**. The carboxyl group of one reacts with the amino group of another, releasing water and forming a peptide linkage.

$H_2N–CH(R_1)–COOH + H_2N–CH(R_2)–COOH$

$\rightarrow H_2N–CH(R_1)–CO–NH–CH(R_2)–COOH + H_2O$

$$H_2N–CH(R_1)–\boxed{CO–NH}–CH(R_2)–\boxed{CO–NH}–CH(R_3)–COOH$$

Peptide linkages: –CO–NH–

Each linkage forms by condensation, releasing H_2O.

Many amino acids join to form a polypeptide chain, which folds into a protein's structure. The peptide linkage is **–CO–NH–**.

Q2(v) Equations: zinc with steam; magnesium with HCl; sodium with cold water.

$Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_2(g)$ [heat]

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

OR — Hydrogen–oxygen fuel cell: reactants, oxidation electrode and overall reaction.

(a) Reactants: hydrogen gas and oxygen gas.

(b) Oxidation occurs at the **anode**, where hydrogen loses electrons.

(c) Overall reaction:

$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

The cell converts the released chemical energy into electrical energy.

Q2(vi) Esterification and why it is a condensation reaction.

Esterification is the reaction of a **carboxylic acid with an alcohol** to form an **ester and water**.

Carboxylic acid + alcohol ⇌ ester + water

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

Ethanoic acid reacts with ethanol to form **ethyl ethanoate and water**. Heat with concentrated sulfuric acid as a catalyst. The reaction is reversible.

Why condensation? Two molecules combine to form a larger molecule while a small molecule of water is eliminated.

General symbolic equation:

$RCOOH + R'OH \rightleftharpoons RCOOR' + H_2O$

OR — How does the salt lattice explain physical properties?

Salts have positive and negative ions arranged in a regular, three-dimensional **giant ionic lattice**.

1. **High melting point:** strong attractions between opposite charges require considerable energy to overcome.

2. **Hard but brittle:** displacement of layers may bring like charges together; repulsion causes the crystal to break.

3. **Conductivity:** solid salts do not conduct because ions are fixed. Molten salts and aqueous solutions conduct because ions can move.

Q2(vii) Complete bromoethane → Y and Y + Br_2 → Z.

$CH_3CH_2Br + KOH \rightarrow CH_2{=}CH_2 + KBr + H_2O$

Conditions: alcoholic KOH and heat. This is elimination (dehydrohalogenation).

Y = ethene, $CH_2{=}CH_2$.

$CH_2{=}CH_2 + Br_2 \rightarrow CH_2Br{-}CH_2Br$

Z = 1,2-dibromoethane. Bromine adds across the double bond.

OR — Why are SO_2/CO_2 acidic and CuO/CaO basic?

SO_2 and CO_2 are **acidic oxides** because they react with bases to form salts and water.

$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

$CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$

CuO and CaO are **basic oxides** because they react with acids to form salts and water.

$CuO + 2HCl \rightarrow CuCl_2 + H_2O$

$CaO + 2HCl \rightarrow CaCl_2 + H_2O$

Q2(viii) Why is petroleum a mixture? Give three hydrocarbons.

Petroleum contains many hydrocarbons that are not chemically combined with one another. Their proportions vary, and they can be separated into fractions by fractional distillation.

Examples: **pentane (C_5H_{12}), hexane (C_6H_{14}), octane (C_8H_{18}).**

OR — Distinguish ethene and ethane using two tests.

Ethene is an **unsaturated alkene** with a C=C bond; ethane is a **saturated alkane** with only single carbon–carbon bonds.

Test 1: bromine water

Add bromine water at room temperature, away from UV light.

Hydrocarbon	Observation	Explanation
Ethene	Decolourises orange-brown bromine water	Bromine adds across the C=C bond
Ethane	No colour change	Saturated; no addition reaction under these conditions

$$CH_2{=}CH_2 + Br_2 \rightarrow CH_2Br{-}CH_2Br$$

The product is **1,2-dibromoethane**.

Test 2: cold dilute alkaline $KMnO_4$

Add **cold, dilute alkaline potassium manganate(VII), $KMnO_4$**, and shake. An alkene removes the purple colour and produces brown manganese dioxide, MnO_2.

$$3CH_2{=}CH_2 + 2KMnO_4 + 4H_2O$$

$$\rightarrow 3HOCH_2CH_2OH + 2MnO_2 + 2KOH$$

Ethene is oxidised to **ethane-1,2-diol**. Ethane gives no colour change under these conditions.

Q2(ix) Reactions showing formation of PAN from unburned hydrocarbons.

Unburned hydrocarbons undergo atmospheric oxidation and can form aldehydes such as ethanal. Sunlight drives the chemistry that generates the radicals involved. The following three reactions lead to PAN:

$$CH_3CHO + OH\bullet \rightarrow CH_3CO\bullet + H_2O$$

$$CH_3CO\bullet + O_2 \rightarrow CH_3C(O)OO\bullet$$

$$CH_3C(O)OO\bullet + NO_2 \rightleftharpoons CH_3C(O)OONO_2$$

The product is **peroxyacetyl nitrate (PAN)**, a component of photochemical smog. It irritates eyes and damages plants. The dot denotes an unpaired electron in a radical.

OR — Name fossil fuels; identify methane-rich and cleanest-burning fuel.

(a) The main fossil fuels are **coal, petroleum and natural gas**.

(b) Natural gas consists mainly of methane, CH_4.

(c) Natural gas is the cleanest-burning of these three: it generally produces less soot, sulfur dioxide and carbon dioxide per unit of energy than coal or petroleum fuels.

Q2(x) Vitamins: sources and importance.

Vitamins are organic nutrients required in small amounts for normal growth, metabolism and good health.

Vitamin	Main sources	Importance
A	Carrots, liver, milk, eggs	Vision, healthy skin and immunity

Vitamin	Main sources	Importance
B group	Whole grains, meat, eggs, pulses	Energy metabolism; several support nerves and red blood cells
C	Citrus fruits, guava, tomatoes	Collagen formation, wound healing and healthy gums
D	Oily fish, egg yolk; also formed in sun-exposed skin	Calcium absorption; strong bones and teeth
E	Vegetable oils, nuts, seeds	Protection against oxidative cell damage
K	Green leafy vegetables	Normal blood clotting

OR — Alcohol combustion and why alcohols are cleaner fuels.

With sufficient oxygen, alcohols burn to form **CO_2, water and heat**.

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O + \text{heat}$$

1. Oxygen in alcohol molecules helps combustion and reduces the tendency to form soot.
2. Pure alcohols contain no sulfur, so they do not form SO_2.
3. Under suitable conditions, they produce less smoke and CO than many petroleum fuels.

Alcohol combustion still releases carbon dioxide.

Q2(xi) Zinc electroplating of iron: procedure and reactions.

1. Clean the iron object to remove rust, dirt and grease.
2. Connect iron to the **negative terminal** of a DC supply: it is the **cathode**.
3. Connect a pure zinc plate to the **positive terminal**: it is the **anode**.
4. Immerse both in an electrolyte containing zinc ions, such as $ZnSO_4$ solution, and pass direct current.

Anode — oxidation:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

Cathode — reduction:

$$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$$

Zinc deposits on iron while the zinc anode dissolves. The coating blocks air and moisture. Zinc also gives **sacrificial protection** because it is more reactive than iron.

OR — Ethanol formation by fermentation and hydration.

(a) Fermentation:

The process is **fermentation**. Glucose is converted into ethanol and carbon dioxide by enzymes in yeast, at approximately **30–35°C**, in the absence of oxygen.

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 \text{ [yeast enzymes]}$$

(b) Catalytic addition of steam:

Ethene reacts reversibly with steam to form ethanol; this is **hydration of ethene**.

$CH_2{=}CH_2(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g)$

Conditions: approximately **300°C**, **60–70 atm**, and a **phosphoric acid (H_3PO_4) catalyst supported on silica**.

SECTION C — Long questions (20 marks)

Q3 Identify functional groups in $HCOOCH_3$, $CH_3–C{\equiv}C–CH_3$, C_3H_6 and $C_4H_{11}OH$.

Given formula	Functional group / intended class
$HCOOCH_3$	Ester linkage –C(=O)–O–; methyl methanoate
$CH_3–C{\equiv}C–CH_3$	Carbon–carbon triple bond; but-2-yne, an alkyne
C_3H_6	Expected: C=C double bond, assuming propene ($CH_3–CH{=}CH_2$)
$C_4H_{11}OH$ (printed)	Intended: hydroxyl group –OH; alcohol

C_3H_6 alone is ambiguous: it can mean propene or cyclopropane, and only propene has C=C. $C_4H_{11}OH$ is not the correct formula of an ordinary saturated monohydric alcohol; likely C_4H_9OH was intended.

OR — Chemical properties of metals with acids, cold water, steam and oxygen.

(i) Dilute acid: a suitable metal above hydrogen reacts with dilute HCl to form a salt and hydrogen.

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

Magnesium dissolves with bubbles of hydrogen.

(ii) Cold water: very reactive metals form a hydroxide and hydrogen.

$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

Sodium moves rapidly on the water surface.

(iii) Steam: suitable heated metals form an oxide and hydrogen.

$Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_2(g)$ [heat]

(iv) Oxygen: many metals form metal oxides.

$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

Magnesium burns with a dazzling white flame and forms white MgO.

Q4 28 g CO reacts with excess O_2: moles of CO and CO_2, mass and RTP volume of CO_2.

$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$

Oxygen is in excess, so CO determines the product amount.

(a) Moles of CO:

$M_r(CO) = 12 + 16 = 28\ g\ mol^{-1}$

$n(CO) = 28 \div 28 = 1.0\ mol$

(b) Moles of CO_2: the ratio $CO:CO_2$ is 2:2 = 1:1.

$n(CO_2) = 1.0$ mol

(c) Mass of CO_2:

$M_r(CO_2) = 44$ g mol^{-1}

$m(CO_2) = 1.0 \times 44 = 44$ g

(d) Volume at RTP:

$V(CO_2) = n \times 24 = 1.0 \times 24 = 24$ dm^3

OR — Empirical/molecular formula: 10.0 g C, 3.0 g H, 2.0 g N; molar mass 70.

The printed data are inconsistent. The calculation below shows why no valid molecular formula can be obtained.

Element	Mass / g	Moles	Divide by 0.142857
C	10.0	10/12 = 0.8333	5.8333
H	3.0	3/1 = 3.000	21
N	2.0	2/14 = 0.142857	1

Multiply the ratio by 6:

C:H:N = 35:126:6

The formal empirical ratio is **$C_{35}H_{126}N_6$**.

Formula mass = 35(12) + 126(1) + 6(14) = 630

Molecular multiplier = 70 ÷ 630 = 1/9

The multiplier must be a positive whole number, so the data require correction. The formal ratio is also inconsistent with ordinary neutral molecular valencies. A hydrocarbon contains only carbon and hydrogen, whereas this question includes nitrogen. Do not round 5.8333 directly to 6 to force an answer.

Q5 Rate of reaction: definition, graph method and interpretation.

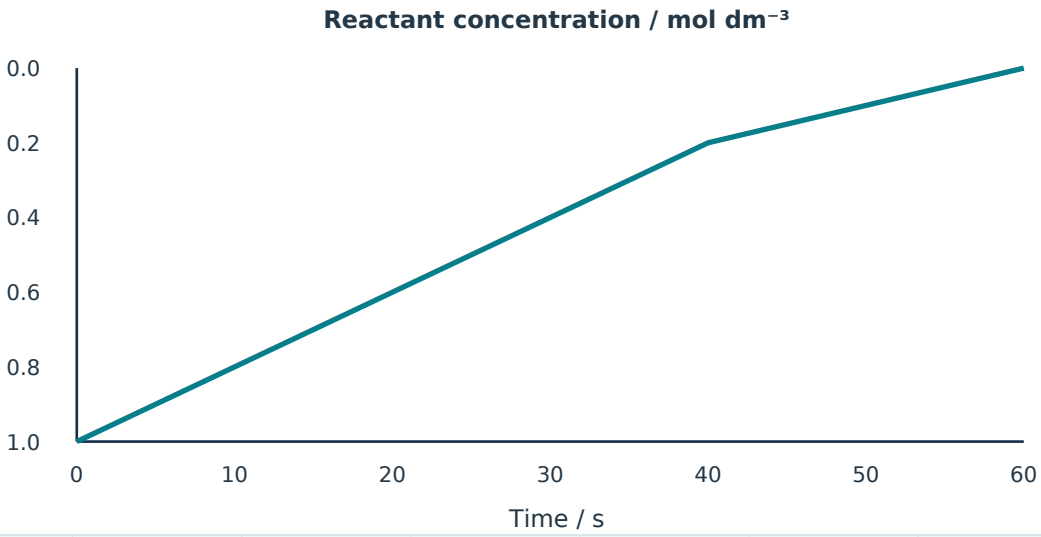
(a) Rate is the change in reactant or product concentration per unit time.

Average rate = $-\Delta[\text{reactant}]/\Delta t = \Delta[\text{product}]/\Delta t$

The negative sign makes the rate positive for a decreasing reactant concentration. Unit: **mol dm^{-3} s^{-1}**. Measure gas volume, mass loss or concentration/colour change at regular intervals.

(b) Plot time horizontally and concentration vertically. Average rate is concentration change divided by time. Instantaneous rate is the magnitude of the tangent gradient. A steeper concentration change means a faster reaction.

(c) Supplied graph: its vertical scale is reversed (1.0 at the bottom, 0.0 at the top), so the rising line shows **decreasing reactant concentration**.



Time / s	0	10	20	30	40	50	60
Concentration / mol dm⁻³	1.0	0.8	0.6	0.4	0.2	0.1	0.0

Rate, 0–40 s = (1.0 − 0.2)/40 = 0.020 mol dm⁻³ s⁻¹

Rate, 40–60 s = (0.2 − 0.0)/20 = 0.010 mol dm⁻³ s⁻¹

The plotted rate is approximately constant initially and lower after 40 s. This is consistent with fewer successful collisions as concentration falls. The plotted reactant is exhausted at about 60 s.

OR — Reactivity of X, Y and Z from water/acid observations.

Decreasing order: X > Y > Z

Metal	Observation	Explanation
X	Releases hydrogen with cold water	Most reactive; displaces hydrogen from water
Y	No reaction with water; releases H_2 with HCl	Less reactive than X, but above hydrogen
Z	No reaction with water or dilute acid	Least reactive; consistent with a metal below hydrogen under ordinary conditions

Possible examples are **X = calcium, Y = zinc, Z = copper**. The observations do not uniquely identify them.

$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$

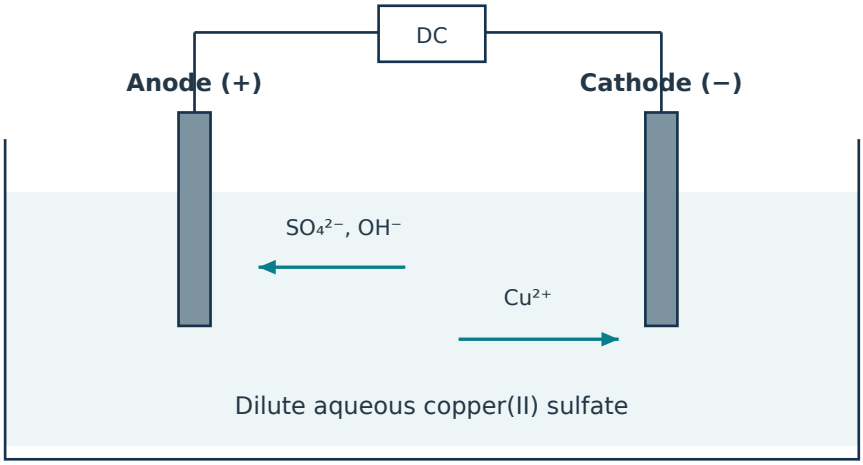
$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Copper does not react with dilute HCl under ordinary conditions.

Q6 Electrolysis of dilute $CuSO_4$: diagram, labels, ions, products and observations.

The question permits inert or copper electrodes. The first answer uses **inert platinum electrodes**.

Use inert electrodes, such as platinum, immersed in dilute aqueous copper(II) sulfate and connected to a DC supply.



Both electrodes immersed; electrodes do not touch

Cathode (negative) — reduction:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

Copper forms a **reddish-brown coating**, increasing the cathode's mass.

Anode (positive) — oxidation:

$$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$$

Equivalent form:

$$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$$

Bubbles of **oxygen** form at the anode. Oxygen relights a glowing splint.

Ion movement: Cu^{2+} and other cations migrate towards the cathode; SO_4^{2-} and OH^- migrate towards the anode. Sulfate ions remain in solution rather than being discharged under these conditions.

Overall observations: the blue colour becomes paler as Cu^{2+} ions are removed; the solution becomes more acidic; copper forms at the cathode and oxygen at the anode.

$$2CuSO_4 + 2H_2O \rightarrow 2Cu + O_2 + 2H_2SO_4$$

Alternative: copper electrodes

Copper electrodes are immersed in aqueous copper(II) sulfate and connected to a DC supply.

Positive copper anode — oxidation:

$$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$$

The anode dissolves and **loses mass**.

Negative copper cathode — reduction:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

A reddish-brown copper deposit forms; the cathode **gains mass**.

Copper is transferred from the anode to the cathode. Ideally, Cu^{2+} concentration and the blue colour remain approximately constant, and anode mass loss equals cathode mass gain. **Oxygen is not the normal anode product with a copper anode.**

OR — Polymer from a dicarboxylic acid and diamine.

A **dicarboxylic acid** (HOOC–R–COOH) and a **diamine** (H_2N–R′–NH_2) form a **polyamide by condensation polymerisation**. A carboxyl group reacts with an amino group, releasing water. Both monomers have two reactive groups, so the chain can continue growing.

–COOH + H_2N– → –CO–NH– + H_2O

General equation, including chain end groups:

n H_2N–R′–NH_2 + n HOOC–R–COOH

→ H[–NH–R′–NH–CO–R–CO–]$_n$OH + (2n − 1)H_2O

$$[–NH–R′–NH–C(=O)–R–C(=O)–]_n$$

Amide linkage: –CO–NH–

R and R′ represent the carbon-containing parts of the monomers.

Functional group: amide linkage, **–CO–NH–**.

Example: nylon-6,6, formed from hexane-1,6-diamine (H_2N–$(CH_2)_6$–NH_2) and hexanedioic acid (HOOC–$(CH_2)_4$–COOH).

Nylon-6,6 repeating unit: [–NH–$(CH_2)_6$–NH–CO–$(CH_2)_4$–CO–]$_n$

Applications: strong fibres for ropes, clothing and carpets.

PAPER 02
1st Annual 2026 — Paper B

Original question paper

The supplied question pages follow. Complete worked solutions begin immediately afterwards.

34

CHEMISTRY SSC–II

Time allowed: 2:40 Hours **Total Marks Sections B and C: 53**

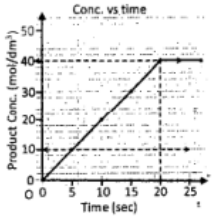
SECTION – B (Marks 33)

Q. 2 Attempt all parts. (11 x 3 = 33) سوال نمبر2 تمام اجزا حل کریں۔

(i)	Given the standard electrode potentials of zinc (–0.76 V) and copper (+0.34 V), identify which metal is more reactive? Justify your answer. زنک (−0.76V) اور کاپر (+0.34V) کے معیاری الیکٹروڈ پوٹینشل دیے گئے ہیں۔ کون سی میٹل زیادہ ری ایکٹو ہے؟ وجہ بھی بیان کریں۔	1+2	OR	Draw the structural and displayed formula of Propan-1-ol پروپین۔1۔اول کا ساختی اور ظاہر شدہ (ڈسپلیڈ) فارمولا بنائیں	1+2
(ii)	Name the catalyst used in the Contact process. Also state its role. کنٹیکٹ پراسس میں استعمال ہونے والے کیٹالسٹ کا نام لکھیں۔ اس کیٹالسٹ کا کردار بھی لکھیں۔	1+2	OR	Calculate the concentration in mol/dm³ of a solution containing 20 g of NaOH in 1 dm³ of solution. NaOH کے 20 گرام $1 dm^3$ سلوشن میں حل پذیر ہیں۔ اس سلوشن کی کنسنٹریشن mol/dm^3 میں معلوم کریں۔	03
(iii)	What are fossil fuels? Name any three fossil fuels. فوسل فیولز کیا ہیں؟ کوئی سے تین فوسل فیولز کے نام لکھیں۔	1.5 + 1.5	OR	State the observation when an unsaturated hydrocarbon reacts with aqueous bromine. Explain with equation. ان سیچوریٹڈ ہائیڈروکاربن کے ایکوس برومین سے ری ایکشن پر کیا آبزرویشن ہوتی ہے؟ ایک مساوات کی مدد سے وضاحت کریں۔	1+2
(iv)	What is a polymer? Give two examples as well. پولیمر کیا ہے؟ دو مثالیں بھی دیں۔	03	OR	Give three applications of electroplating. الیکٹروپلیٹنگ کی تین ایپلیکیشنز تحریر کریں۔	1x3
(v)	State what happens when a reactive metal reacts with a dilute acid. Give an example of reaction with balanced equation. ایک زی ایکٹو میٹل جب کسی ڈائیلوٹ ایسڈ کے ساتھ ری ایکٹ کرے تو کیا ہوتا ہے؟ ایک مثال دیں اور اس کی متوازن مساوات بھی لکھیں۔	1+2	OR	How can ethene (C_2H_4) be prepared from: ایتھین (C_2H_4) کو درج شدہ سے کیسے بنایا جا سکتا ہے؟ a Bromoethane بروموایتھین b Ethanol ایتھانول	03
(vi)	What is esterification? Give an example having balanced chemical equation. ایسٹری فیکیشن کیا ہے؟ ایک مثال دیں اور متوازن کیمیائی مساوات لکھیں۔	1x3	OR	What is the usual state of ionic compounds under normal conditions? Give two reasons. آئیونک مرکبات نارمل کنڈیشن میں عام طور پر کس حالت میں پائے جاتے ہیں؟ دو وجوہات بھی لکھیں۔	1+2
(vii)	Enlist any three advantages of Hydrogen-Oxygen fuel cell. ہائیڈروجن آکسیجن فیول سیل کے کوئی سے تین فوائد لکھیں۔	1x3	OR	Is calcium oxide (CaO) considered a basic oxide? Justify with chemical equation. کیا کیلشیم آکسائیڈ کو بیسک آکسائیڈ سمجھا جاتا ہے؟ کیمیائی مساوات سے جسٹیفائی کریں۔	1+2
(viii)	What is petroleum? Why does it need to be separated into fractions? پیٹرولیم کیا ہے؟ اسے مختلف حصوں میں تقسیم کی کیوں ضرورت ہوتی ہے؟	1+2	OR	The empirical formula of a compound is CH_2O. Find its molecular formula if molar mass is 180. ایک کمپاؤنڈ کا امپیریکل فارمولا CH_2O ہے۔ اگر اس کا مولر ماس 180 ہو تو اس کا مالیکیولر فارمولا معلوم کریں۔	03
(ix)	How nitrogen oxides react with unburned hydrocarbons to form PAN? Give three reactions. نائٹروجن آکسائیڈز غیر جلے ہوئے ہائیڈروکاربن کے ساتھ ری ایکشن کر کے PAN کیسے بناتے ہیں؟ تین مساواتیں لکھیں۔	1x3	OR	Calculate the concentration in g/dm³ of a solution that contains 0.25 mol of KNO_3 in 1 dm^3 of solution. سلوشن کے ایک dm^3 میں KNO_3 کے 0.25 مول موجود ہیں۔ اس کی کنسنٹریشن g/dm^3 میں معلوم کریں۔	03
(x)	Name two sources of vitamin A and state one of its functions. وٹامن A کے دو ذرائع بتائیں اور اس کا ایک فعل بیان کریں۔	1x3	OR	Write equation of complete combustion of alcohols. Write names of products formed. الکوحل کے مکمل احتراق (کمبسچن) کی مساوات لکھیں۔ نیز پراڈکٹس کے نام لکھیں۔	2+1
(xi)	How amino acids join together to form a protein? Show with an equation. امینو ایسڈز کس پروٹین کیسے بناتے ہیں؟ مساوات کی مدد سے دکھائیں۔	03	OR	Write the complete balanced chemical equation for the addition of steam to ethene to form ethanol. ایتھین میں بھاپ کے اضافے سے ایتھانول بنانے کی مکمل متوازن کیمیائی مساوات لکھیں۔	03

SECTION – C (Marks 20)

Note: Attempt all questions. تمام سوالات حل کریں۔

Q.3	Enlist any four petroleum fractions. Also given their uses. پیٹرولیم فریکشن کے کسی بھی چار حصوں کے نام لکھیں اور ان کے استعمالات بھی بیان کریں۔	1x4	OR	How do the metals react with the following: میٹلز درج شدہ کے ساتھ کیسے ری ایکٹ کرتی ہیں؟ a. Steam (سٹیم (بھاپ)) b. oxygen (آکسیجن) (Include one relevant specific chemical equation each) (ہر ایک کی ایک متعلقہ کیمیائی مساوات بھی شامل کریں۔)	2+2
Q.4	Classify the following oxides as acidic, basic or amphoteric based on their chemical properties: SO_2, CuO, Al_2O_3 (Give balanced chemical equations) درج آکسائیڈز کو ان کی کیمیائی خصوصیات کی بنیاد پر بطور ایسڈک بیسک یا ایمفوٹرک درجہ بندی کریں Al_2O_3، CuO، SO_2 (متوازن کیمیائی مساوات بھی لکھیں)	2x3	OR	In given reaction, if 150g Zn reacts with 200g dil. H_2SO_4. (Mr. Zn=63.5 $gmol^{-1}$) Determine: درج مساوات میں Zn کے 150g کے ساتھ H_2SO_4 کے 200g کا ری ایکشن ہو تو معلوم کریں۔ مولر ماس زنک = $63.5 gmol^{-1}$ $Zn + H_2SO_4 \longrightarrow 2ZnSO_4 + H_2$ a. Limiting Reactant (لیمیٹنگ ری ایکٹنٹ) b. Mass of H_2 produced (بننے والی H_2 کا ماس)	6
Q.5	C= 38.71%, H = 9.68% O = 51.61% Calculate the empirical and molecular formulas of given compound from the given data if its Molar Mass is 62 g/mol. دی گئی معلومات سے مذکورہ کمپاؤنڈ کا مالیکیولر اور امپیریکل فارمولا نکالیں جبکہ (مولر ماس: 62g/mol)	06	OR	a. What does the rate of reaction mean? How it can be measured? b. Calculate the rate of reaction between 5 and 20 sec in given graph. c. What does the horizontal line at the end of graph indicate? Explain.  الف۔ ریٹ آف ری ایکشن سے کیا مراد ہے؟ اسے کیسے ناپا جاتا ہے؟ ب۔ دیے گئے گراف میں 5 سے 20 سیکنڈ کے درمیان ریٹ آف ری ایکشن معلوم کریں۔ ج۔ گراف کے آخر میں افقی لکیر کیا ظاہر کرتی ہے؟ وضاحت کریں۔	2x3
Q.6	What happens during the electrolysis of dilute copper sulphate using copper electrodes? Also give the equations of chemical reactions occurring at both electrodes. ڈائلیوٹ کاپر سلفیٹ کے محلول کی الیکٹرالسس جب کاپر کے الیکٹروڈز کے ذریعے کی جاتی ہے تو کیا عمل ہوتا ہے؟ اس پر اس میں ہونے والے ری ایکشنز کی کیمیائی مساواتیں دیں۔	2+2	OR	Describe the formation of a polyamide from a dicarboxylic acid and a diamine. Include a general equation and the type of linkage formed. ڈائیکاربوکسیلک ایسڈ اور ڈائی امین سے پولی امائیڈ کی تشکیل بیان کریں۔ عمومی مساوات اور بننے والی لنکیج کی قسم بھی تحریر کریں۔	2+2

—— 2SA-I 26005 (B) ——

SUPPLEMENTARY TABLE

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40

Chemistry page 2 of 2

1st Annual 2026 — Paper B

Complete solutions • Sections B and C

SECTION B — Q2: Short questions (33 marks)

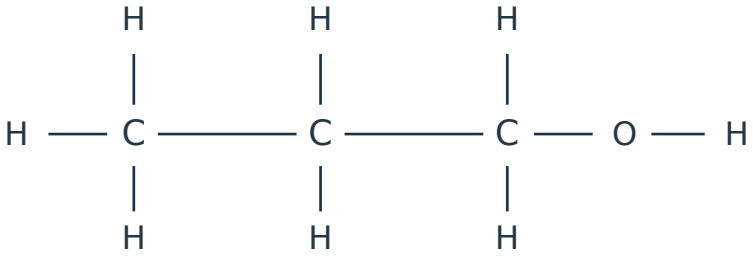
Q2(i) Which is more reactive: Zn (−0.76 V) or Cu (+0.34 V)?

Zinc is more reactive. Its standard reduction potential is more negative, so it loses electrons and undergoes oxidation more readily than copper.

$Zn \rightarrow Zn^{2+} + 2e^-$

OR — Structural and displayed formula of propan-1-ol.

$CH_3–CH_2–CH_2–OH$



Q2(ii) Catalyst in the Contact Process and its role.

The catalyst is **vanadium(V) oxide, V_2O_5**. It provides an alternative reaction pathway with lower activation energy, increasing the rate of SO_2 conversion to SO_3. It remains chemically unchanged overall and does not change the equilibrium position.

$2SO_2 + O_2 \rightleftharpoons 2SO_3$

OR — Concentration of 20 g NaOH in 1 dm³ of solution.

$M_r(NaOH) = 23 + 16 + 1 = 40 \text{ g mol}^{-1}$

$n = m/M_r = 20/40 = 0.50 \text{ mol}$

$c = n/V = 0.50/1 = 0.50 \text{ mol dm}^{-3}$

Q2(iii) What are fossil fuels? Give three examples.

Fossil fuels are **non-renewable fuels** formed from the remains of ancient organisms through heat and pressure over millions of years. Three examples are **coal, petroleum and natural gas**.

OR — Observation and equation: unsaturated hydrocarbon with aqueous bromine.

The orange-brown colour disappears as bromine adds across a carbon–carbon multiple bond. For ethene:

$CH_2=CH_2 + Br_2 \rightarrow CH_2Br–CH_2Br$

The product is **1,2-dibromoethane**. This is an **addition reaction**.

Q2(iv) Define polymer and give two examples.

A polymer is a very large molecule formed by repeatedly joining small molecules called **monomers** into a chain.

Examples: **polyethene**, from ethene; **PVC (polyvinyl chloride)**, from chloroethene.

OR — Three applications of electroplating.

1. Chromium plating of taps and vehicle parts gives a shiny, corrosion-resistant surface.
2. Silver plating of cutlery improves appearance.
3. Zinc plating of iron protects it from corrosion.

Q2(v) Reactive metal with a dilute acid: result and balanced equation.

A suitable metal above hydrogen reacts with dilute HCl or dilute H_2SO_4 to produce a **salt and hydrogen gas**.

Metal + dilute acid → salt + hydrogen

$Mg + 2HCl \rightarrow MgCl_2 + H_2$

Magnesium dissolves and bubbles of hydrogen are released.

OR — Prepare ethene from bromoethane and ethanol.

(a) Bromoethane: heat with alcoholic KOH; dehydrohalogenation (elimination).

$CH_3CH_2Br + KOH \rightarrow CH_2{=}CH_2 + KBr + H_2O$

(b) Ethanol: heat with concentrated H_2SO_4 at about 170°C; dehydration.

$CH_3CH_2OH \rightarrow CH_2{=}CH_2 + H_2O$

Q2(vi) Esterification: definition and balanced example.

Esterification is the reaction of a **carboxylic acid with an alcohol** to form an **ester and water**.

Carboxylic acid + alcohol ⇌ ester + water

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

Ethanoic acid reacts with ethanol to form **ethyl ethanoate and water**. Heat with concentrated sulfuric acid as a catalyst. The reaction is reversible.

OR — Usual state of ionic compounds and two reasons.

Ionic compounds are generally **crystalline solids** under normal conditions.

1. Oppositely charged ions are held by strong electrostatic attractions.
2. The ions form a rigid giant lattice requiring considerable energy to break apart.

Q2(vii) Three advantages of a hydrogen–oxygen fuel cell.

1. Its reaction product is **water**, so it produces no CO_2 or soot during operation.
2. It converts chemical energy directly into electrical energy with **high efficiency**.
3. The fuel-cell stack operates quietly, with no moving mechanical parts.

OR — Is CaO basic? Justify with an equation.

Yes. CaO neutralises acids, forming salt and water.

$$CaO + 2HCl \rightarrow CaCl_2 + H_2O$$

It also reacts with water to form calcium hydroxide, an alkali.

$$CaO + H_2O \rightarrow Ca(OH)_2$$

Q2(viii) What is petroleum and why separate it into fractions?

Petroleum (crude oil) is a naturally occurring mixture consisting mainly of hydrocarbons. Components have different boiling points and different useful properties. Fractional distillation separates fractions such as petrol for cars, kerosene for aircraft and bitumen for roads.

OR — Empirical formula CH_2O; molar mass 180. Find molecular formula.

Empirical formula mass = 12 + 2(1) + 16 = 30

n = 180/30 = 6

Molecular formula = $(CH_2O)_6 = C_6H_{12}O_6$

Q2(ix) Nitrogen oxides, unburned hydrocarbons and PAN: three reactions.

Unburned hydrocarbons undergo atmospheric oxidation and can form aldehydes such as ethanal. Sunlight drives the chemistry that generates the radicals involved. The following three reactions lead to PAN:

$$CH_3CHO + OH\bullet \rightarrow CH_3CO\bullet + H_2O$$

$$CH_3CO\bullet + O_2 \rightarrow CH_3C(O)OO\bullet$$

$$CH_3C(O)OO\bullet + NO_2 \rightleftharpoons CH_3C(O)OONO_2$$

The product is **peroxyacetyl nitrate (PAN)**, a component of photochemical smog. It irritates eyes and damages plants. The dot denotes an unpaired electron in a radical.

OR — Concentration in g/dm³ of 0.25 mol KNO_3 in 1 dm³.

$M_r(KNO_3) = 39 + 14 + 3(16) = 101 \text{ g mol}^{-1}$

$m = nM_r = 0.25 \times 101 = 25.25 \text{ g}$

Mass concentration = $m/V = 25.25/1 = 25.25 \text{ g dm}^{-3}$

Q2(x) Two sources and one function of vitamin A.

Sources: carrots (beta-carotene, converted by the body into vitamin A) and liver.

Function: normal vision, especially in dim light. Deficiency may cause **night blindness**.

OR — Complete combustion of alcohols and products.

In sufficient oxygen, alcohols burn to form **carbon dioxide and water**, releasing heat. For ethanol:

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$$

Q2(xi) How amino acids form proteins, with an equation.

Amino acids join through **condensation reactions**. The carboxyl group of one reacts with the amino group of another, releasing water and forming a peptide linkage.

$H_2N–CH(R_1)–COOH + H_2N–CH(R_2)–COOH$

$\rightarrow H_2N–CH(R_1)–CO–NH–CH(R_2)–COOH + H_2O$

$H_2N–CH(R_1)–\boxed{CO–NH}–CH(R_2)–\boxed{CO–NH}–CH(R_3)–COOH$

Peptide linkages: –CO–NH–

Each linkage forms by condensation, releasing H_2O.

Many amino acids join to form a polypeptide chain, which folds into a protein's structure. The peptide linkage is **–CO–NH–**.

OR — Complete balanced equation for steam addition to ethene.

Ethene reacts reversibly with steam to form ethanol; this is **hydration of ethene**.

$CH_2=CH_2(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g)$

Conditions: approximately **300°C**, **60–70 atm**, and a **phosphoric acid (H_3PO_4) catalyst supported on silica**.

SECTION C — Long questions (20 marks)

Q3 Four petroleum fractions and their uses.

Fraction	Use
Refinery gases	Bottled gas for cooking and heating
Petrol / gasoline	Fuel for cars
Kerosene	Jet aircraft fuel
Diesel / gas oil	Fuel for diesel engines, buses and trucks

OR — Metals with steam and oxygen, including equations.

(a) Steam: suitable heated metals form a metal oxide and hydrogen.

$Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_2(g)$ [heat]

(b) Oxygen: many metals form metal oxides.

$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

Magnesium burns with a dazzling white flame, forming white MgO.

Q4 Classify SO_2, CuO and Al_2O_3 with balanced equations.

SO_2 is acidic: it reacts with a base to form a salt and water.

$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

CuO is basic: it reacts with an acid to form a salt and water.

$CuO + 2HCl \rightarrow CuCl_2 + H_2O$

Al_2O_3 is amphoteric: it reacts with both acids and bases.

$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$

$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$

The second reaction uses hot aqueous sodium hydroxide and forms sodium tetrahydroxoaluminate.

OR — 150 g Zn reacts with 200 g dilute H_2SO_4: limiting reactant and H_2 mass.

The printed equation has an incorrect coefficient before $ZnSO_4$. The correct equation is $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$. The expected calculation assumes 200 g is the mass of H_2SO_4 solute. If 200 g means dilute solution, its concentration is required.

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

Use the zinc molar mass supplied in the question, 63.5 g mol^{-1}.

$n(Zn) = 150/63.5 = 2.362$ mol

$n(H_2SO_4) = 200/98 = 2.041$ mol

The ratio $Zn:H_2SO_4$ is 1:1. Fewer moles of acid are present, so **H_2SO_4 is the limiting reactant** under the stated assumption.

$n(H_2) = 2.041$ mol

$m(H_2) = 2.041 \times 2 = 4.08$ g (approximately)

Q5 Empirical and molecular formula: C 38.71%, H 9.68%, O 51.61%; molar mass 62.

Assume 100 g of compound.

Element	Moles	Divide by smallest
C	38.71/12 = 3.226	1.00
H	9.68/1 = 9.68	3.00
O	51.61/16 = 3.226	1.00

C:H:O = 1:3:1

Empirical formula: CH_3O.

Empirical formula mass = 12 + 3 + 16 = 31

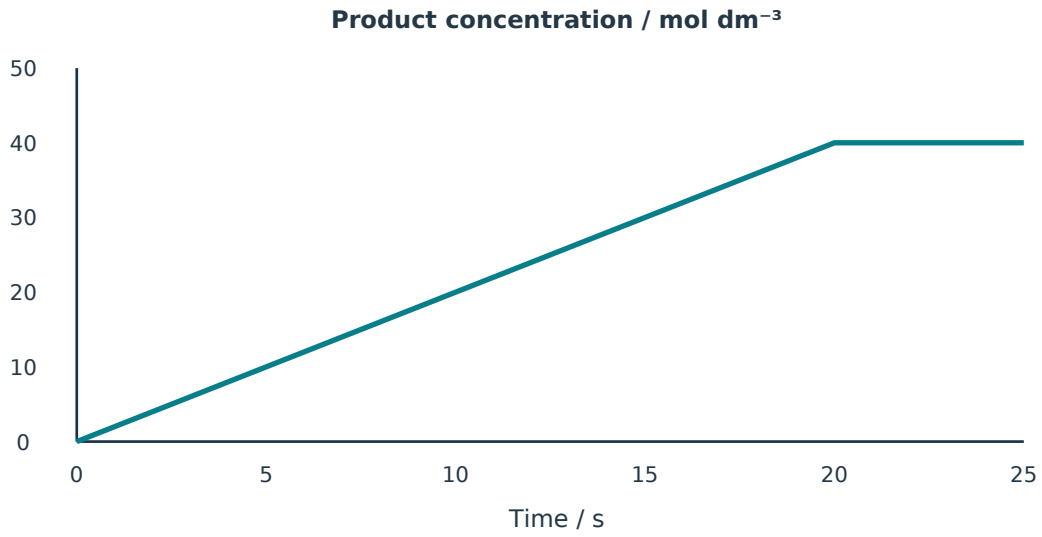
n = 62/31 = 2

Molecular formula: $C_2H_6O_2$.

OR — Rate definition, measurement, graph calculation and plateau.

(a) Reaction rate is change in reactant or product concentration per unit time. Measure gas volume, mass loss or concentration change at regular time intervals.

Rate = change in concentration / time taken



(b) At 5 s, product concentration = 10 mol dm⁻³; at 20 s it is 40 mol dm⁻³.

Rate = (40 − 10)/(20 − 5) = 30/15

Rate = 2.0 mol dm⁻³ s⁻¹

(c) The horizontal line means concentration is constant, no additional product is forming and measured rate is zero. In the intended interpretation, the reaction has finished because a limiting reactant is used up.

Q6 Electrolysis of dilute $CuSO_4$ using copper electrodes.

Copper electrodes are immersed in aqueous copper(II) sulfate and connected to a DC supply.

Positive copper anode — oxidation:

$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$

The anode dissolves and **loses mass**.

Negative copper cathode — reduction:

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

A reddish-brown copper deposit forms; the cathode **gains mass**.

Copper is transferred from the anode to the cathode. Ideally, Cu^{2+} concentration and the blue colour remain approximately constant, and anode mass loss equals cathode mass gain. **Oxygen is not the normal anode product with a copper anode.**

OR — Formation of a polyamide: general equation and linkage.

A **dicarboxylic acid** (HOOC–R–COOH) and a **diamine** (H_2N–R′–NH_2) form a **polyamide by condensation polymerisation**. A carboxyl group reacts with an amino group, releasing water. Both monomers have two reactive groups, so the chain can continue growing.

–COOH + H_2N– → –CO–NH– + H_2O

General equation, including chain end groups:

n H_2N–R′–NH_2 + n HOOC–R–COOH

→ H[–NH–R′–NH–CO–R–CO–]$_n$OH + (2n − 1)H_2O

$$[–NH–R′–NH–C(=O)–R–C(=O)–]_n$$

Amide linkage: –CO–NH–

R and R′ represent the carbon-containing parts of the monomers.

Functional group: amide linkage, **–CO–NH–**.

Example: nylon-6,6, formed from hexane-1,6-diamine ($H_2N–(CH_2)_6–NH_2$) and hexanedioic acid ($HOOC–(CH_2)_4–COOH$).

Nylon-6,6 repeating unit: $[–NH–(CH_2)_6–NH–CO–(CH_2)_4–CO–]_n$

Applications: strong fibres for ropes, clothing and carpets.

PAPER 03
1st Annual 2026 — Paper D

Original question paper

The supplied question pages follow. Complete worked solutions begin immediately afterwards.

36

CHEMISTRY SSC–II

Time allowed: 2:40 Hours **Total Marks Sections B and C: 53**

SECTION – B (Marks 33)

Q. 2 Attempt all parts. (11 x 3 = 33) سوال نمبر2 تمام اجزا حل کریں۔

(i)	Given the standard electrode potentials of aluminum (–1.67 V) and copper (+0.34 V), identify which metal is more reactive? Justify your answer. ایلومینیم (−1.67V) اور کاپر (+0.34V) کے معیاری الیکٹروڈ پوٹینشل دیے گئے ہیں۔ کون سی میٹل زیادہ ری ایکٹو ہے؟ وجہ بھی بیان کریں۔	1+2	OR	Draw the structural and displayed formula of butan-1-ol. بیوٹین-1-اول کا ساختی اور ظاہر شدہ (ڈسپلیڈ) فارمولا بنائیں۔	1+2
(ii)	Write complete balanced chemical equation for the conversion of sulfur dioxide to sulfur trioxide in the Contact process. کانٹیکٹ پراسس میں سلفر ڈائی آکسائیڈ کو سلفر ٹرائی آکسائیڈ میں تبدیل کرنے کی مکمل اور متوازن کیمیائی مساوات لکھیں۔	03	OR	Calculate the concentration in mol/dm³ of a solution containing 10 g of NaCl in 1 dm³ of solution. 10 گرام NaCl 1dm³ سلوشن میں حل پذیر ہیں۔ اس سلوشن کی کنسنٹریشن mol/dm³ میں معلوم کریں۔	03
(iii)	Write balanced chemical equation of complete combustion of Methanol. Write names of products. میتھانول کے مکمل احتراق (کمبسچن) کی مساوات لکھیں۔ نیز پراڈکٹس کے نام بھی لکھیں۔	2+1	OR	How can presence of double bond be identified in hydrocarbons? Give chemical reaction. کسی ہائیڈروکاربن میں ڈبل بانڈ کی موجودگی کیسے معلوم کی جاسکتی ہے؟ کیمیائی مساوات لکھیں۔	2+1
(iv)	What is meant by the term monomer? Give two examples. مونومر سے کیا مراد ہے؟ دو مثالیں بھی دیں۔	2+1	OR	What is electroplating? Give its two uses. الیکٹروپلیٹنگ کیا ہے؟ اس کے دو استعمالات لکھیں۔	03
(v)	What type of compound is formed when metals react with oxygen? Give one chemical equation. جب دھاتیں آکسیجن کے ساتھ ری ایکشن کرتی ہیں تو کس قسم کا مرکب بنتا ہے؟ ایک کیمیائی مساوات دیں۔	2+1	OR	State three advantages of a hydrogen–oxygen fuel cell over fossil-fuel. ہائیڈروجن–آکسیجن فیول سیل کے تین فوائد بیان کریں۔ جو فوسل فیول میں نہیں۔	1x3
(vi)	Give balanced chemical reaction between ethanoic acid and methanol. Also write name of the products formed. ایتھانوئک ایسڈ اور میتھانول کے ری ایکشن کی متوازن مساوات لکھیں۔ نیز بننے والی پراڈکٹس کے نام تحریر کریں۔	1+2	OR	Why do ionic compounds have high melting points? Give two reasons. آئنک مرکبات کا میلٹنگ پوائنٹ زیادہ کیوں ہوتا ہے؟ دو وجوہات لکھیں۔	1.5 + 1.5
(vii)	Write the preparation of C_3H_6 (propene) from: پروپین (C_3H_6) کی درج شدہ سے تیاری لکھیں۔ a Bromo-propane بروموپروپین b Propane-1-ol پروپین-1- اول	03	OR	Is CO_2 considered an acidic oxide? Justify with chemical reaction. کیا CO_2 کو ایسڈک آکسائیڈ سمجھا جاتا ہے؟ ایک مساوات کی مدد سے جسٹیفائی کریں۔	1+2
(viii)	What is meant by fractional distillation? Enlist any two petroleum fractions. فریکشنل ڈسٹلیشن سے کیا مراد ہے؟ کوئی سی دو پیٹرولیم فریکشنز کے نام لکھیں۔	2+1	OR	The empirical formula of a compound is CH_2O. Find its molecular formula if molar mass is 180. ایک کمپاؤنڈ کا امپیریکل فارمولا CH_2O ہے۔ اگر اس کا مولر ماس 180 ہو تو اس کا مالیکیولر فارمولا معلوم کریں۔	03
(ix)	What is Haber process? Write its balanced chemical equation with conditions. ہیبر کا پراسس کیا ہے؟ اس کی متوازن مساوات کنڈیشنز کے ساتھ لکھیں۔	1+2	OR	What is fossil fuel? Name any three fossil fuels. فوسل فیول کیا ہے؟ کوئی سے تین فوسل فیولز کے نام لکھیں۔	1.5 + 1.5
(x)	Name two sources of vitamin D and state one of its functions. وٹامن D کے دو ذرائع بتائیں اور اس کا ایک فعل بیان کریں۔	1x3	OR	Calculate the concentration in mol/dm³ of a solution containing 9.8 g of (H_2SO_4) in 0.5 dm³ of solution. ایک سلوشن کے $0.5dm^3$ میں H_2SO_4 کے 9.8g موجود ہیں۔ اس کی کنسنٹریشن mol/dm³ میں معلوم کریں۔	03
(xi)	Draw the general structure of an amino acid and label its main functional groups. امینو ایسڈ کی عمومی ساخت بنائیں اور اس کے اہم فنکشنل گروپس کی نشاندہی کریں۔	1+2	OR	State one advantage and one disadvantage of fermentation compared to the catalytic hydration of ethene. ایتھین کی کیٹلٹک ہائیڈریشن کے مقابلے میں خمیر سازی کا ایک فائدہ اور ایک نقصان بیان کریں۔	1.5 + 1.5

SECTION – C (Marks 20)

Note: Attempt all questions. تمام سوالات حل کریں۔

Q.3	Describe of the uses of ethyne in: ایتھائن کے درج شدہ استعمالات لکھیں۔ **a** Artificial ripening of fruits (Reaction of CaC_2 and water) — پھلوں کو مصنوعی طور پر پکانا (CaC_2 اور پانی کا ری ایکشن) **b** Welding — ویلڈنگ	2+2	OR	Discuss the reactivity of following metals with oxygen. درج میٹلز کی آکسیجن کے ساتھ ری ایکٹویٹی بیان کریں: **a.** Na **b.** Al **c.** Cu **d.** Pt	1x4
Q.4	In given reaction, if 68.1g $CaCO_3$ reacts with 51.6g HCl. Determine: 68.1g $CaCO_3$ کے 51.6g HCl کے ساتھ کاری ایکشن درج ایکشن میں ہو تو معلوم کریں۔ $CaCO_3 + 2HCl \longrightarrow CaCl_2 + CO_2 + H_2O$ **a** Limiting Reactant — لیمیٹنگ ری ایکٹنٹ **b** Mass of $CaCl_2$ produced — بننے والے $CaCl_2$ کا ماس	3+3	OR	Classify the following oxides as acidic, basic or amphoteric based on their chemical properties: SO_2, CuO, Al_2O_3 Give balanced chemical equations درج شدہ آکسائیڈز کی ان کی کیمیائی خصوصیات کی بنیاد پر بطور ایسڈک، بیسک یا ایمفوٹیرک درجہ بندی کریں SO_2، CuO، Al_2O_3 متوازن کیمیائی مساوات بھی لکھیں۔	2x3
Q.5	Highlight the Role of chemical kinetics in food industry. (Any four aspects) فوڈ انڈسٹر میں کیمیکل کائینٹکس کے کردار پر تفصیلی روشنی ڈالیں۔ (کوئی سے چار پہلو)	1.5x4	OR	C= 40%, H = 6.67% O = 53.33% Calculate the empirical and molecular formulas of given compound from the given data if its Molar Mass is 90 g/mol. دی گئی معلومات سے مذکورہ کمپاؤنڈ کا مالیکیولر اور امپیریکل فارمولا نکالیں جبکہ (مولر ماس: 90g/mol)	06
Q.6	Compare the electrolysis of any halide compound in dilute and concentrated solutions. کسی ہیلائیڈ کمپاؤنڈ کی ڈائلیوٹ اور کنسنٹریٹڈ سلوشنز میں الیکٹرالسس کا موازنہ کریں۔	2+2	OR	Describe the differences between addition polymer ization and condensation polymerization. (Two points each.) ایڈیشن پولیمرائزیشن اور کنڈینسیشن پولیمرائزیشن کے درمیان فرق بیان کریں۔ (دو دو نکات لکھیں۔)	2x4

SUPPLEMENTARY TABLE

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40

1st Annual 2026 — Paper D

Complete solutions • Sections B and C

SECTION B — Q2: Short questions (33 marks)

Q2(i) Which is more reactive: Al (−1.67 V) or Cu (+0.34 V)?

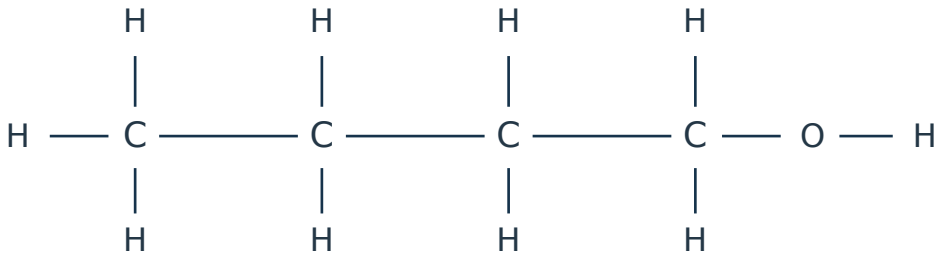
Aluminium is more reactive. It has the more negative standard reduction potential, so it loses electrons more readily.

$$Al \rightarrow Al^{3+} + 3e^-$$

Aluminium's protective oxide layer can slow observed reactions, but the metal itself is more reactive than copper.

OR — Structural and displayed formula of butan-1-ol.

$CH_3–CH_2–CH_2–CH_2–OH$



Q2(ii) Balanced Contact Process conversion of SO_2 to SO_3.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

Conditions: approximately **450°C, 1–2 atm, V_2O_5 catalyst**.

OR — Concentration of 10 g NaCl in 1 dm^3 of solution.

$M_r(NaCl) = 23 + 35.5 = 58.5$ g mol^{-1}

$n = 10/58.5 = 0.17094$ mol

$c = n/V = 0.17094/1 = 0.171$ mol dm^{-3} (approximately)

The correct relative atomic mass of sodium is 23. The supplementary table appears to misprint it.

Q2(iii) Complete combustion of methanol and products.

$$2CH_3OH + 3O_2 \rightarrow 2CO_2 + 4H_2O$$

Products: carbon dioxide and water.

OR — Identify a hydrocarbon double bond by chemical test.

Add bromine water at room temperature, away from UV light. The orange-brown colour disappears because bromine adds across the C=C bond.

$$CH_2=CH_2 + Br_2 \rightarrow CH_2Br–CH_2Br$$

Q2(iv) Define monomer and give two examples.

A monomer is a small molecule that can join repeatedly with other monomers to form a polymer. Examples: **ethene** (forms polyethene) and **chloroethene** (forms PVC).

OR — Define electroplating and give two uses.

Electroplating deposits a thin layer of one metal on an object by passing current through a suitable electrolyte.

1. Corrosion protection, for example zinc plating of iron.

2. Improved appearance, for example silver plating of cutlery.

Q2(v) What forms when metals react with oxygen? Give an equation.

Metals generally form **metal oxides**. Many metal oxides are basic; some, including Al_2O_3, are amphoteric.

$$2Mg + O_2 \rightarrow 2MgO$$

OR — Three advantages of a hydrogen–oxygen fuel cell over fossil fuel use.

1. Its reaction product is **water**, so it produces no CO_2 or soot during operation.

2. It converts chemical energy directly into electrical energy with **high efficiency**.

3. The fuel-cell stack operates quietly, with no moving mechanical parts.

Q2(vi) Ethanoic acid + methanol: equation and product names.

$$CH_3COOH + CH_3OH \rightleftharpoons CH_3COOCH_3 + H_2O$$

Heat with concentrated H_2SO_4 as a catalyst. Products are **methyl ethanoate (an ester)** and **water**.

OR — Two reasons for high melting points of ionic compounds.

1. Oppositely charged ions are held together by **strong electrostatic attractions** throughout a giant ionic lattice.

2. Considerable heat energy is required to overcome these attractions so that ions can move past one another during melting. Therefore, ionic compounds have high melting points.

Q2(vii) Prepare propene from bromopropane and propan-1-ol.

(a) Heat 1-bromopropane with alcoholic KOH (dehydrohalogenation).

$$CH_3CH_2CH_2Br + KOH \rightarrow CH_3CH{=}CH_2 + KBr + H_2O$$

(b) Heat propan-1-ol with concentrated H_2SO_4 (dehydration).

$$CH_3CH_2CH_2OH \rightarrow CH_3CH{=}CH_2 + H_2O$$

OR — Is CO_2 acidic? Justify with reaction.

Yes. It reacts with a base to form a salt and water.

$$CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$$

It also forms weak carbonic acid in water:

$$CO_2 + H_2O \rightleftharpoons H_2CO_3$$

Q2(viii) Define fractional distillation and name two fractions.

Fractional distillation separates a liquid mixture into fractions according to different boiling points, through repeated vaporisation and condensation in a fractionating column.

Examples: **petrol and kerosene**.

OR — Empirical formula CH_2O; molar mass 180. Find molecular formula.

Empirical formula mass = 12 + 2 + 16 = 30

n = 180/30 = 6

Molecular formula = $C_6H_{12}O_6$

Q2(ix) Haber Process: definition, equation and conditions.

The Haber Process is the industrial manufacture of **ammonia from nitrogen and hydrogen**.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Conditions: approximately **450°C**, approximately **200 atm**, finely divided **iron catalyst**, and nitrogen:hydrogen mole ratio **1:3**.

OR — Fossil fuel definition and three examples.

Fossil fuels are **non-renewable fuels** formed from the remains of ancient organisms through heat and pressure over millions of years. Three examples are **coal, petroleum and natural gas**.

Q2(x) Two sources and one function of vitamin D.

Dietary sources: oily fish and egg yolk. Vitamin D is also made in skin exposed to sunlight.

Function: helps calcium absorption and maintains strong bones and teeth. Deficiency can cause **rickets in children**.

OR — Concentration of 9.8 g H_2SO_4 in 0.5 dm^3 solution.

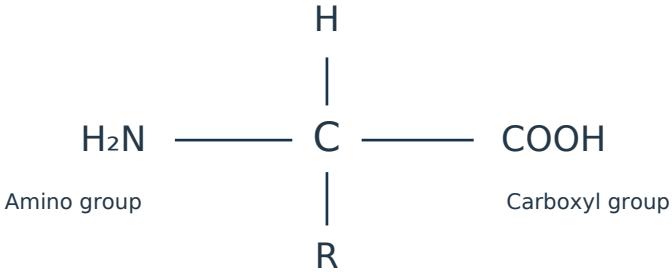
$M_r(H_2SO_4) = 98\ g\ mol^{-1}$

n = 9.8/98 = 0.10 mol

$c = 0.10/0.5 = 0.20\ mol\ dm^{-3}$

Q2(xi) General amino acid structure and functional groups.

$H_2N–CH(R)–COOH$



R = variable side chain

–NH_2: amino group. **–COOH:** carboxyl group. **R:** variable side chain. The central carbon is also bonded to a hydrogen atom.

OR — One advantage and disadvantage of fermentation compared with hydration.

Advantage: uses renewable plant-derived sugars and runs at low temperature and normal pressure.

Disadvantage: slower; produces dilute, impure ethanol that requires purification. Catalytic hydration is faster and can run continuously.

SECTION C — Long questions (20 marks)

Q3 Ethyne in artificial fruit ripening and welding.

(a) Fruit ripening: calcium carbide reacts with water to generate ethyne (acetylene).

$$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$$

Acetylene can stimulate ripening. This explains the textbook use described in the question.

Carbide-based fruit ripening is unsafe. Ethylene is the appropriate commercial ripening gas; the reaction above is included to explain the examination question.

(b) Welding: ethyne burns in oxygen to give a very hot oxyacetylene flame (approximately 3,000–3,500°C), used for welding and cutting metals.

$$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$$

OR — Reactivity of Na, Al, Cu and Pt with oxygen.

Sodium: reacts rapidly. Simplified sodium oxide formation:

$$4Na + O_2 \rightarrow 2Na_2O$$

In excess oxygen it can form sodium peroxide:

$$2Na + O_2 \rightarrow Na_2O_2$$

Aluminium: forms a protective surface oxide layer, preventing further rapid oxidation underneath.

$$4Al + 3O_2 \rightarrow 2Al_2O_3$$

Copper: on heating, forms black copper(II) oxide.

$$2Cu + O_2 \rightarrow 2CuO \text{ [heat]}$$

Platinum: very unreactive; does not react appreciably with oxygen under ordinary conditions.

Q4 68.1 g $CaCO_3$ + 51.6 g HCl: limiting reactant and mass of $CaCl_2$.

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$$

$$M_r(CaCO_3) = 40 + 12 + 48 = 100 \text{ g mol}^{-1}$$

$$n(CaCO_3) = 68.1/100 = 0.681 \text{ mol}$$

$$n(HCl) = 51.6/36.5 = 1.414 \text{ mol}$$

HCl required for 0.681 mol carbonate:

$$2 \times 0.681 = 1.362 \text{ mol HCl}$$

Available HCl (1.414 mol) exceeds the requirement. **$CaCO_3$ is the limiting reactant.**

$$n(CaCl_2) = 0.681 \text{ mol}$$

$$M_r(CaCl_2) = 40 + 2(35.5) = 111 \text{ g mol}^{-1}$$

$m(CaCl_2) = 0.681 \times 111 = 75.591\ g \approx 75.6\ g$

OR — Classify SO_2, CuO and Al_2O_3 with equations.

SO_2 is acidic: it reacts with a base to form a salt and water.

$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

CuO is basic: it reacts with an acid to form a salt and water.

$CuO + 2HCl \rightarrow CuCl_2 + H_2O$

Al_2O_3 is amphoteric: it reacts with both acids and bases.

$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$

$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$

The second reaction uses hot aqueous sodium hydroxide and forms sodium tetrahydroxoaluminate.

Q5 Four roles of chemical kinetics in the food industry.

Chemical kinetics studies reaction rates and factors affecting them.

1. Preservation and storage: refrigeration slows many spoilage reactions and microbial processes, extending storage life.

2. Shelf-life prediction: measured rates of oxidation, nutrient loss and other changes help estimate how long food remains safe and acceptable.

3. Cooking and heat treatment: rate studies help choose cooking, pasteurisation and sterilisation times and temperatures, balancing microbial destruction against flavour and nutrient loss.

4. Fermentation control: temperature, pH and other conditions are adjusted to control production of bread, yoghurt and other fermented foods.

OR — Empirical/molecular formula: C 40%, H 6.67%, O 53.33%; molar mass 90.

Assume 100 g.

Element	Moles	Simplest ratio
C	40/12 = 3.333	1
H	6.67/1 = 6.67	2
O	53.33/16 = 3.333	1

Empirical formula: CH_2O.

Empirical formula mass = 12 + 2 + 16 = 30

n = 90/30 = 3

Molecular formula: $C_3H_6O_3$.

Q6 Compare electrolysis of a halide in dilute and concentrated solutions.

Use **aqueous sodium chloride** with inert electrodes.

Dilute NaCl — cathode: water gains electrons to form hydrogen.

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

Dilute NaCl — anode: in sufficiently dilute solution, oxygen is the expected main product.

$$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$$

Overall dilute solution:

$$2H_2O \rightarrow 2H_2 + O_2$$

Concentrated NaCl (brine) — cathode:

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$$

Concentrated NaCl — anode:

$$2Cl^- \rightarrow Cl_2 + 2e^-$$

Overall, with products kept separate:

$$2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$$

Solution	Cathode product	Main anode product
Dilute NaCl	Hydrogen	Oxygen
Concentrated NaCl	Hydrogen	Chlorine

The key difference is the **anode product**.

OR — Addition versus condensation polymerisation.

Feature	Addition	Condensation
Monomers	Usually unsaturated, containing C=C	Two reactive functional groups per monomer
Small molecule eliminated?	No	Yes, commonly water
Example	Ethene → polyethene	Diamine + dicarboxylic acid → polyamide

Addition:

$$n\ CH_2{=}CH_2 \rightarrow [{-}CH_2{-}CH_2{-}]_n$$

Condensation linkage formation:

$${-}COOH + H_2N{-} \rightarrow {-}CO{-}NH{-} + H_2O$$

PAPER 04
1st Annual 2026 — Paper M

Original question paper

The supplied question pages follow. Complete worked solutions begin immediately afterwards.

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CHEMISTRY SSC–II

Time allowed: 2:40 Hours **Total Marks Sections B and C: 53**

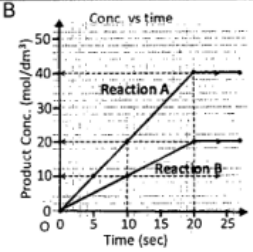
SECTION – B (Marks 33)

Q. 2 Attempt all parts. (11 x 3 = 33) سوال نمبر2 تمام اجزا حل کریں۔

(i)	What are amphoteric oxides? Give two examples. ایمفوٹیرک آکسائیڈ کیا ہوتے ہیں؟ دو مثالیں دیں۔	1+2	OR	The empirical formula of a compound is CH_2O. Find its molecular formula if molar mass is 60. ایک کمپاؤنڈ کا ایمپیریکل فارمولا CH_2O ہے۔ اگر اس کا مولر ماس 60 ہو تو اس کا مالیکیولر فارمولا معلوم کریں۔	03
(ii)	Briefly describe the following: درج شدہ کی مختصر وضاحت کریں: a Monomer مونومر b Polymer پولی مر	1.5 + 1.5	OR	Write the chemical reactions occurring in a hydrogen–oxygen fuel cell. Also give reaction at anode and cathode ہائیڈروجن۔آکسیجن فیول سیل میں اینوڈ اور کیتھوڈ پر ہونے والے ری ایکشنز کی مساوات بھی نیز مجموعی ری ایکشن بھی لکھیں۔	1x3
(iii)	Give two sources of vitamin C and mention one deficiency disease caused by its lack. وٹامن سی کے دو ذرائع لکھیں اور اس کی کمی سے ہونے والی ایک بیماری کا نام بتائیں۔	1x3	OR	Write the balanced chemical equation for the complete combustion of ethanol. ایتھانول کے مکمل احتراق (کمبسجن) کی متوازن کیمیائی مساوات لکھیں۔	03
(iv)	Write the general word equation for the esterification reaction. Also give a specific example. ایسٹری فیکیشن کی عمومی لفظی مساوات لکھیں۔ ایک مثال بھی دیں۔	1+2	OR	Why do ionic compounds have high melting points? Give two reasons. آئنک مرکبات کا میلٹنگ پوائنٹ زیادہ کیوں ہوتا ہے؟ دو وجوہات لکھیں۔	03
(v)	Write balanced chemical equation for following processes: درج شدہ پراسسز کے لیے متوازن کیمیائی مساوات تحریر کریں: a Dehydration of Ethanol ایتھانول کی ڈی ہائیڈریشن b Dehydrohalogenation of chloroethane کلوروایتھین کی ڈی ہائیڈروہیلوجنیشن	03	OR	In a reaction between magnesium and hydrochloric acid, 1.2 g of magnesium reacts with excess acid. Calculate the mass and moles of hydrogen gas produced. میگنیشیم اور ہائیڈروکلورک ایسڈ کے ری ایکشن میں 1.2g میگنیشیم ہائیڈروکلورک ایسڈ کی وافر مقدار سے ری ایکٹ کر رہا ہے۔ بننے والی ہائیڈروجن گیس کا ماس اور مول کی تعداد معلوم کریں۔	03
(vi)	What is meant by fractional distillation? Enlist any two petroleum fractions. فریکشنل ڈسٹلیشن سے کیا مراد ہے؟ کوئی سی دو پیٹرولیم فریکشنز کے نام لکھیں۔	1+2	OR	Through which process glucose is converted into ethanol? Write balance chemical equation. کس پراسس سے گلوکوز کو ایتھانول میں تبدیل کیا جاتا ہے؟ متوازن کیمیائی مساوات لکھیں۔	1+2
(vii)	What is Haber process? Write its balanced chemical equation with conditions. ہیبر کا پراسس کیا ہے؟ اس کی متوازن مساوات کنڈیشنز کے ساتھ لکھیں۔	03	OR	Briefly state any two uses of alcohols in daily life. روزمرہ زندگی میں الکحل کوئی سے دو استعمالات مختصراً بیان کریں۔	1.5 + 1.5
(viii)	What happens when a metal reacts with steam? Give an example of reaction with balanced chemical equation. جب کوئی میٹل بھاپ کے ساتھ ری ایکشن کرے تو کیا ہوتا ہے؟ ایک مثال دیں اور متوازن کیمیائی مساوات لکھیں۔	03	OR	Draw the general structure of a dipeptide formed from two amino acids. دو امینو ایسڈز سے بننے والے ڈائی پیپٹائیڈ کی عمومی ساخت بنائیں۔	03
(ix)	Which metal is commonly used for chrome plating? Give its chemical reactions at anode and cathode. کروم پلیٹنگ کے لیے عام طور پر کون سی دھات استعمال کی جاتی ہے؟ اس میں اینوڈ اور کیتھوڈ پر ہونے والی ری ایکشنز کی مساواتیں بھی لکھیں۔	03	OR	Describe the potassium manganate (VII) ($KMnO_4$) test for unsaturated hydrocarbons with balanced equation. غیر سیر شدہ ہائیڈروکاربن کے لیے پوٹاشیم مینگنیٹ (VII) ٹیسٹ بیان کریں متوازن مساوات بھی دیں۔	03
(x)	Sketch a schematic diagram of Daniel cell. ڈینیئل سیل کی سکیمیٹک ڈائیگرام بنائیں۔	03	OR	Draw the structural and displayed formula of Butan-2-ol بیوٹین-2-اول کا ساختی اور ظاہر شدہ (ڈسپلیڈ) فارمولا بنائیں۔	1+2
(xi)	State the optimum conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process. کنٹیکٹ پراسس میں سلفر ڈائی آکسائیڈ کو سلفر ٹرائی آکسائیڈ میں تبدیل کرنے کے لیے موزوں حالات بیان کریں۔	1x3	OR	Calculate the concentration in mol/dm³ of a solution containing 10 g of Na_2CO_3 in 1 dm³ of solution. Na_2CO_3 کے 10 گرام $1 dm^3$ سلوشن میں حل پذیر ہیں۔ اس سلوشن کی کنسنٹریشن mol/dm^3 میں معلوم کریں۔	03

SECTION – C (Marks 20)

Note: Attempt all questions. تمام سوالات حل کریں۔

Q.3	Give comparison of hydrogen oxygen fuel cells with gasoline/petrol engines. (Four points) ہائیڈروجن آکسیجن فیول سیلز اور گیسولین/پیٹرول انجنوں کا موازنہ پیش کریں۔ (چار نکات)	04	OR	Write balance reactions of Metals with (one each): درج شدہ کے ساتھ میٹلز کا ایک ایک متوازن ری ایکشن لکھیں۔ a – Cold water – ٹھنڈا پانی b – Dilute acids – ڈائلیوٹ ایسڈز	2+2
Q.4	In given reaction, if 159g Na_2CO_3 reacts with 73g HCl. Determine: درج ایکشن میں 159g کے Na_2CO_3 کے ساتھ 73g کے HCl کاری ایکشن ہو تو معلوم کریں۔ $Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$ a – Limiting Reactant – لیمیٹنگ ری ایکٹنٹ b – Mass of NaCl produced – بننے والے NaCl کا ماس	06	OR	Classify CO_2, CaO and Al_2O_3 as acidic, basic or amphoteric oxides based on their chemical properties. Give balanced chemical equations آکسائیڈز CO_2, CaO اور Al_2O_3 کی ان کی کیمیائی خصوصیات کی بنیاد پر ایسڈک، بیسک یا ایمفوٹیرک درجہ بندی کریں۔ متوازن کیمیائی مساوات بھی لکھیں۔	2x
Q.5	Two reactions A and B are given? a. Which reaction is faster? Given reason with calculated values. b. What does the horizontal line at the end of graph indicate? Explain.  دو ری ایکشنز A اور B دیے گئے ہیں۔ الف۔ دونوں ری ایکشنز میں سے کون سا تیز ہے؟ کیلکیولیٹڈ ویلیوز کے ساتھ اس کی وجہ بھی بتائیں۔ ب۔ گراف کے آخر میں افقی لکیر کیا ظاہر کرتی ہے؟ وضاحت کریں۔	4+2	OR	C= 54.55%, H = 9.09% O = 36.36% Calculate the empirical and molecular formulas of given compound from the given data if its Molar Mass is 88 g/mol. دی گئی معلومات سے مذکورہ کمپاؤنڈ کا مالیکیولر اور امپیریکل فارمولا نکالیں جبکہ (مولر ماس: 88g/mol)	06
Q.6	Bottom to top in fractionating column, discuss the change in following properties of petroleum fractions in detail. فریکشنیٹنگ کالم میں نیچے سے اوپر درج شدہ خصوصیات میں تبدیلی تفصیلاً بیان کریں۔ a – Chain length – چین لینتھ b – Boiling point – بوائلنگ پوائنٹ	2+2	OR	Describe the environmental challenges. Caused by plastic, limited to: درج نکات کی روشنی میں پلاسٹک کی وجہ سے پیدا ہونے والے ماحولیاتی مسائل کی وضاحت کریں۔ a – Disposal in landfill sites – لینڈ فل سائٹس میں نکلی b – Accumulation in oceans – سمندروں میں جمع ہونا	2+2

SUPPLEMENTARY TABLE

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40

1st Annual 2026 — Paper M

Complete solutions • Sections B and C

SECTION B — Q2: Short questions (33 marks)

Q2(i) Amphoteric oxides and two examples.

Amphoteric oxides react with **both acids and bases**, forming salts and water or corresponding complex salts. Examples: **Al_2O_3** and **ZnO**.

OR — Empirical formula CH_2O; molar mass 60. Find molecular formula.

Empirical formula mass = 12 + 2 + 16 = 30

n = 60/30 = 2

Molecular formula = $C_2H_4O_2$

Q2(ii) Briefly describe monomer and polymer.

Monomer: a small molecule that can repeatedly join others to form a polymer; example, ethene.

Polymer: a large molecule consisting of many repeating units derived from monomers; example, polyethene.

$n\ CH_2{=}CH_2 \rightarrow [–CH_2–CH_2–]_n$

OR — Hydrogen–oxygen fuel cell: anode, cathode and overall reactions.

For a hydrogen–oxygen fuel cell with an **alkaline electrolyte**:

Anode — oxidation:

$2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$

Cathode — reduction:

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

Overall:

$2H_2 + O_2 \rightarrow 2H_2O$

Electrons flow from the anode to the cathode through the external circuit. Chemical energy is converted into electrical energy.

Q2(iii) Two sources and deficiency disease of vitamin C.

Sources: citrus fruits (oranges and lemons) and guava.

Deficiency: scurvy, which can cause bleeding gums and poor wound healing.

OR — Complete combustion of ethanol.

$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

Carbon dioxide and water form, and heat is released.

Q2(iv) Esterification: general word equation and example.

Esterification is the reaction of a **carboxylic acid with an alcohol** to form an **ester and water**.

Carboxylic acid + alcohol ⇌ ester + water

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

Ethanoic acid reacts with ethanol to form **ethyl ethanoate and water**. Heat with concentrated sulfuric acid as a catalyst. The reaction is reversible.

OR — Two reasons for high melting points of ionic compounds.

1. Oppositely charged ions are held together by **strong electrostatic attractions** throughout a giant ionic lattice.

2. Considerable heat energy is required to overcome these attractions so that ions can move past one another during melting. Therefore, ionic compounds have high melting points.

Q2(v) Dehydration of ethanol and dehydrohalogenation of chloroethane.

(a) Ethanol: concentrated H_2SO_4, approximately 170°C.

$CH_3CH_2OH \rightarrow CH_2=CH_2 + H_2O$

(b) Chloroethane: alcoholic KOH and heat.

$CH_3CH_2Cl + KOH \rightarrow CH_2=CH_2 + KCl + H_2O$

OR — Mass and moles of H_2 from 1.2 g Mg with excess HCl.

$Mg + 2HCl \rightarrow MgCl_2 + H_2$

$n(Mg) = 1.2/24 = 0.050$ mol

The Mg:H_2 mole ratio is 1:1.

$n(H_2) = 0.050$ mol

$m(H_2) = nM_r = 0.050 \times 2 = 0.10$ g

Q2(vi) Fractional distillation and two petroleum fractions.

Fractional distillation separates a liquid mixture into fractions with different boiling points through repeated vaporisation and condensation. Examples: **petrol and diesel**.

OR — Process converting glucose into ethanol; balanced equation.

The process is **fermentation**. Glucose is converted into ethanol and carbon dioxide by enzymes in yeast, at approximately **30–35°C**, in the absence of oxygen.

$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ [yeast enzymes]

Q2(vii) Haber Process: definition, equation and conditions.

The Haber Process is the industrial manufacture of **ammonia from nitrogen and hydrogen**.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Conditions: approximately **450°C**, approximately **200 atm**, finely divided **iron catalyst**, and nitrogen:hydrogen mole ratio **1:3**.

OR — Two daily-life uses of alcohol.

1. **Solvents:** ethanol is used in perfumes, cosmetics and some medicines.
2. **Fuels:** ethanol is used directly as fuel and blended with petrol.

Q2(viii) Metal with steam: result and balanced example.

Suitable metals react with steam on heating to form a **metal oxide and hydrogen**.

$$Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_2(g)\ [heat]$$

Not all metals react with steam; copper does not under ordinary laboratory conditions.

OR — General dipeptide structure formed from two amino acids.

$$H_2N–CH(R_1)–CO–NH–CH(R_2)–COOH$$

Peptide linkage: –CO–NH–. One water molecule is released when the two amino acids join.

$$H_2N–CH(R_1)–COOH + H_2N–CH(R_2)–COOH$$

$$\rightarrow H_2N–CH(R_1)–CO–NH–CH(R_2)–COOH + H_2O$$

Q2(ix) Metal used in chrome plating and electrode reactions.

The coating metal is **chromium (Cr)**. The object being plated is the **cathode**.

Simplified school model with chromium(III) ions and a soluble chromium anode:

$$\text{Anode: } Cr(s) \rightarrow Cr^{3+}(aq) + 3e^-$$

$$\text{Cathode: } Cr^{3+}(aq) + 3e^- \rightarrow Cr(s)$$

Chromium deposits on the object, giving a hard, shiny, corrosion-resistant surface.

Technical clarification: conventional industrial chromium plating commonly uses an insoluble lead-alloy anode, not a dissolving chromium anode. The equations above are a simplified soluble-anode model. Actual cathode chemistry depends on the chromium bath.

The main oxygen-evolution reaction at a conventional insoluble anode is:

$$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$$

OR — Potassium manganate(VII) test for unsaturated hydrocarbons.

Add **cold, dilute alkaline potassium manganate(VII), $KMnO_4$**, and shake. An alkene removes the purple colour and produces brown manganese dioxide, MnO_2.

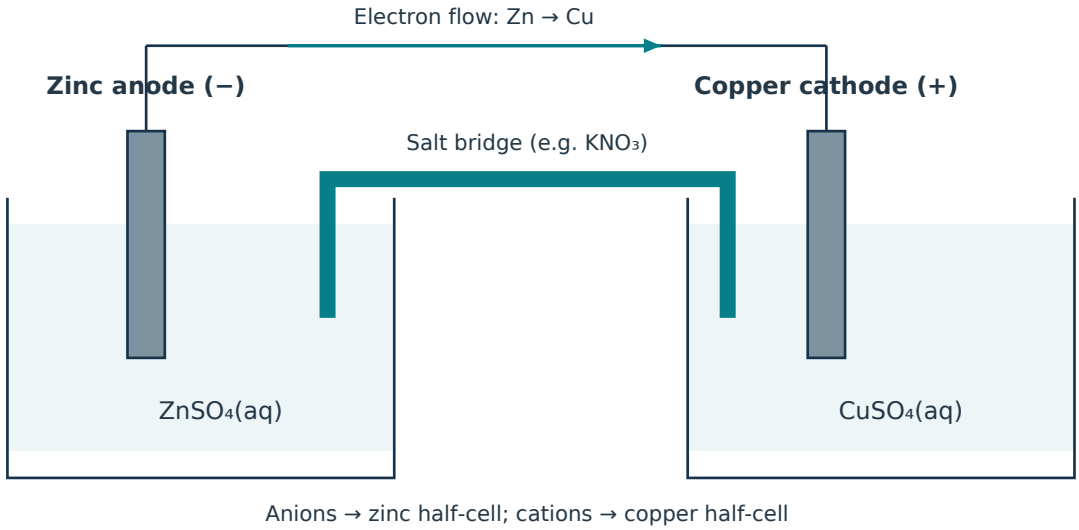
$$3CH_2{=}CH_2 + 2KMnO_4 + 4H_2O$$

$$\rightarrow 3HOCH_2CH_2OH + 2MnO_2 + 2KOH$$

Ethene is oxidised to **ethane-1,2-diol**. Ethane gives no colour change under these conditions.

Q2(x) Sketch and label a Daniell cell.

A Daniell cell has a zinc electrode in $ZnSO_4$ solution and a copper electrode in $CuSO_4$ solution. The electrodes are joined by an external wire and the solutions by a salt bridge.



Anode (negative), oxidation:

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Cathode (positive), reduction:

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Overall:

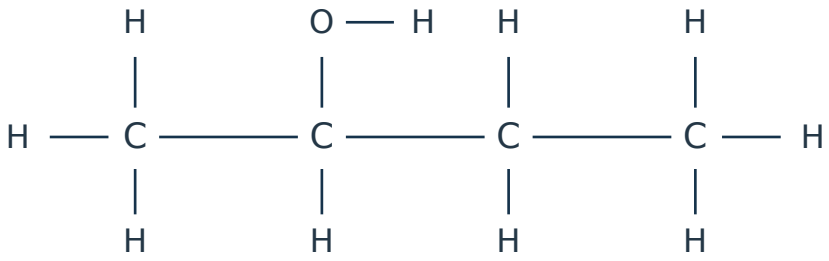
$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

Electrons flow **from zinc to copper** through the external wire. The salt bridge completes the circuit and maintains electrical neutrality: anions migrate towards the zinc half-cell and cations towards the copper half-cell.

Cell notation: $Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$

OR — Structural and displayed formula of butan-2-ol.

$CH_3–CH(OH)–CH_2–CH_3$



Q2(xi) Optimum conditions for $SO_2 \rightarrow SO_3$ in the Contact Process.

$2SO_2 + O_2 \rightleftharpoons 2SO_3$

Temperature: approximately 450°C. **Pressure:** approximately 1–2 atm. **Catalyst:** vanadium(V) oxide, V_2O_5.

OR — Concentration of 10 g Na_2CO_3 in 1 dm^3 of solution.

$M_r(Na_2CO_3) = 2(23) + 12 + 3(16) = 106$

$n = 10/106 = 0.09434$ mol

$c = n/V = 0.09434/1 \approx 0.0943 \text{ mol dm}^{-3}$

SECTION C — Long questions (20 marks)

Q3 Compare hydrogen–oxygen fuel cells and petrol engines in four points.

Feature	Hydrogen–oxygen fuel cell	Petrol engine
Conversion	Chemical energy directly to electrical energy	Chemical energy to heat, then mechanical energy
Efficiency	Generally higher conversion efficiency	Substantial energy loss as heat
Products in operation	Water	CO_2 and water; may also emit CO, NO_x and unburned hydrocarbons
Noise/moving parts	Quiet stack with no moving mechanical parts	Many moving parts; noise and vibration

OR — Balanced reactions of metals with cold water and dilute acids.

(a) Cold water: very reactive metals form a metal hydroxide and hydrogen.

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

Sodium moves rapidly on water and releases hydrogen.

(b) Dilute acid: a suitable metal above hydrogen forms salt and hydrogen with dilute HCl.

$Mg + 2HCl \rightarrow MgCl_2 + H_2$

Magnesium dissolves with bubbles of hydrogen.

Q4 159 g Na_2CO_3 + 73 g HCl: limiting reactant and NaCl mass.

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$

$n(Na_2CO_3) = 159/106 = 1.50 \text{ mol}$

$n(HCl) = 73/36.5 = 2.00 \text{ mol}$

For 1.50 mol Na_2CO_3, HCl required = 1.50 × 2 = 3.00 mol. Only 2.00 mol is available. **HCl is the limiting reactant.**

The HCl:NaCl ratio is 2:2 = 1:1.

$n(NaCl) = 2.00 \text{ mol}$

$m(NaCl) = 2.00 \times 58.5 = 117 \text{ g}$

OR — Classify CO_2, CaO and Al_2O_3 with equations.

CO_2: acidic, reacts with bases.

$CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$

CaO: basic, reacts with acids.

$CaO + 2HCl \rightarrow CaCl_2 + H_2O$

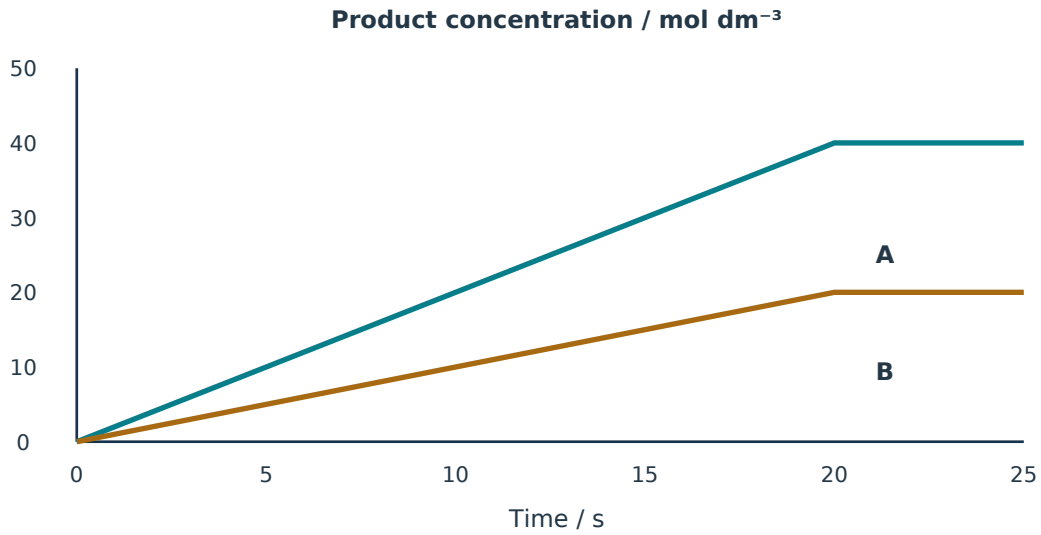
Al_2O_3: amphoteric, reacts with acids and bases.

$$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$$

$$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$$

The alkali reaction uses hot aqueous sodium hydroxide.

Q5 Compare rates of reactions A and B using the graph.



(a) Reaction A is faster: its graph has a steeper gradient. Use 0–20 s.

$$\text{Rate A} = (40 - 0)/(20 - 0) = 2.0 \text{ mol dm}^{-3} \text{ s}^{-1}$$

$$\text{Rate B} = (20 - 0)/(20 - 0) = 1.0 \text{ mol dm}^{-3} \text{ s}^{-1}$$

A forms product twice as fast as B over the straight, rising portions shown.

(b) Horizontal sections mean constant product concentration, no further product formation and zero measured rate. In the intended interpretation, the reactions have finished because their limiting reactants have been used up.

OR — Empirical/molecular formula: C 54.55%, H 9.09%, O 36.36%; molar mass 88.

Assume 100 g.

Element	Moles	Divide by 2.2725
C	54.55/12 = 4.546	≈ 2
H	9.09/1 = 9.09	4
O	36.36/16 = 2.2725	1

Empirical formula: C_2H_4O.

$$\text{Empirical formula mass} = 2(12) + 4(1) + 16 = 44$$

$$n = 88/44 = 2$$

Molecular formula: $C_4H_8O_2$.

Q6 Bottom-to-top changes in petroleum fraction chain length and boiling point.

(a) Chain length decreases upwards. Long-chain hydrocarbons condense near the hot bottom. Shorter-chain molecules rise further and condense nearer the cooler top; the smallest may leave as gases.

(b) Boiling point decreases upwards. Larger molecules have stronger intermolecular attractions and need more energy to separate. Smaller molecules have weaker attractions and lower boiling points. The column is hot at the bottom and cooler at the top, so fractions condense at different heights.

Property	Bottom	Top
Chain length	Longer	Shorter
Boiling point	Higher	Lower
Example	Heavy fuel oil / residue	Refinery gases

OR — Environmental problems: plastic in landfill and oceans.

(a) Landfill: most conventional plastics are not readily biodegradable, so they persist and occupy space. They may fragment into microplastics; some additives can enter surrounding soil and water. Disposal also wastes useful material and manufacturing resources.

(b) Oceans: animals may swallow plastics, causing injury, blockage, starvation or death; become entangled in bags, ropes or fishing nets; and encounter microplastics that spread through marine environments and enter food webs.

PAPER 05
2nd Annual 2026 — Chemistry SSC-II

Original question paper

The supplied question pages follow. Complete worked solutions begin immediately afterwards.

CHEMISTRY SSC–II

10240920260052205

Time allowed: 2:40 Hours **Total Marks Sections B and C: 53**

SECTION – B (Marks 33)

Q. 2 Attempt all parts. **(11 x 3 = 33)** سوال نمبر2 تمام اجزا حل کریں۔

(i)	Arrange the following metals in order of increasing reactivity by considering given standard reduction potential values. $Mg^{2+} + 2e^- \rightarrow Mg$ $E°= -2.37$ $Cu^{2+} + 2e^- \rightarrow Cu$ $E°= +0.34$ $Ni^{2+} + 2e^- \rightarrow Ni$ $E°= -0.25$ دی گئی سٹنڈرڈ ریڈکشن پوٹینشل کی ویلیوز کو مدِ نظر رکھتے ہوئے درج دھاتوں کو بڑھتی ہوئی تعامل پذیری کی ترتیب میں لکھیں۔	03	OR	Write the chemical equations for the reactions of: a. Zinc with steam. b. Magnesium with hydrochloric acid. c. Sodium with cold water. درج تعاملات کی کیمیائی مساوات لکھیں: الف۔ زنک کا بھاپ کے ساتھ تعامل ب۔ میگنیشیم کا ہائیڈروکلورک ایسڈ کے ساتھ تعامل ج۔ سوڈیم کا ٹھنڈے پانی کے ساتھ تعامل	1+1 +1
(ii)	Interpret heating and cooling curves in terms of kinetic theory? حرکی نظریہ کی روشنی میں گرم کرنے اور ٹھنڈا کرنے کے منحنی خطوط کی تشریح کریں۔	1.5 + 1.5	OR	What is the molarity of a solution made by dissolving 98 g of H_2SO_4 in 1 dm^3 of water? (Molar mass of H_2SO_4 = 98 g/mol) 98 g سلفیورک ایسڈ (H_2SO_4) کو 1dm^3 پانی میں حل کر کے بنائے گئے محلول کی مولیریٹی کیا ہوگی؟	03
(iii)	A sample of magnesium carbonate ($MgCO_3$) contains 15 grams of magnesium. What is the mass of carbon dioxide (CO_2) produced on thermal decomposition of magnesium carbonate? میگنیشیم کاربونیٹ ($MgCO_3$) کے ایک نمونے میں 15 گرام میگنیشیم موجود ہے۔ میگنیشیم کاربونیٹ کے حرارتی تجزیے سے پیدا ہونے والی کاربن ڈائی آکسائیڈ (CO_2) کی کمیت معلوم کریں۔	03	OR	Draw structural formula of following: a. But-2-ene b. Prop-1-ol c. Ethanoic acid درج ذیل کے ساختی فارمولے بنائیں: الف۔ بیوٹ-2-ائن (But-2-ene) ب۔ پروپ-1-اول (Prop-1-ol) ج۔ ایتھانوئک ایسڈ	1+1 +1
(iv)	Sketch a schematic diagram for Daniel cell labeling anode, cathode and direction of flow of electron. ڈینیئل سیل (Daniell Cell) کا خاکہ بنائیں اور اس میں اینوڈ، کیتھوڈ اور الیکٹرانوں کے بہاؤ کی سمت کی نشاندہی کریں۔	1+1 +1	OR	Draw the structure showing how protein molecule is formed, clearly highlighting the peptide linkage within the structure. پروٹین کے سالمے کی تشکیل کو ظاہر کرنے والی ساخت بنائیں اور اس ساخت میں پیپٹائڈ روابط (Peptide Linkages) کو واضح طور پر نمایاں کریں۔	2+1
(v)	Calculate moles of oxygen gas having a volume of 25 dm^3 at room temperature? کمرے کے درجہ حرارت پر 25dm^3 حجم رکھنے والی آکسیجن گیس کے مولز کی تعداد معلوم کریں۔	03	OR	Write chemical reactions at anode and cathode along with the overall reaction equation for a hydrogen-oxygen fuel cell. ہائیڈروجن-آکسیجن ایندھن سیل کے لیے اینوڈ اور کیتھوڈ پر ہونے والے کیمیائی تعاملات کے ساتھ مجموعی تعامل کی مساوات بھی لکھیں۔	1+1 +1
(vi)	Compare hydration and fermentation for the formation of ethanol. ایتھانول کی تیاری کے لیے ہائیڈریشن اور فرمنٹیشن کا موازنہ کریں۔	1.5 + 1.5	OR	Enlist any three general solubility rules for salts. نمکیات کی حل پذیری کے کوئی تین عمومی اصول درج کریں۔	03
(vii)	Complete the following reaction equations: a. $CH_3\text{-}CH_2\text{-}Br$ + alcoholic KOH → b. $H_2C{=}CH_2$ + Br_2 → درج تعاملات مکمل کریں:	1.5 + 1.5	OR	Justify that SO_2 is acidic oxides and CaO is basic oxide by giving one for each suitable chemical reaction equation? موزوں کیمیائی تعامل کی ایک ایک مساوات دے کر ثابت کریں کہ SO_2 ایک تیزابی آکسائیڈ جبکہ CaO ایک اساسی آکسائیڈ ہے۔	1.5 + 1.5
(viii)	Enlist any six fractions of a hydrocarbon found in petroleum. پیٹرولیم میں پائے جانے والے ہائیڈروکاربن کے کوئی چھ اجزاء (Fractions) درج کریں۔	0.5x6	OR	Distinguish between Ethene and ethane by giving a chemical test. ایتھین (Ethene) اور ایتھین (Ethane) میں فرق بیان کریں اور اپنے جواب کی تائید کے لیے موزوں کیمیائی ٹیسٹ دیں۔	1.5 + 1.5
(ix)	Illustrate the role of nitrogen oxides and unburned hydrocarbons towards the formation of peroxyacetyl nitrates (PANs) by giving suitable chemical equations? موزوں کیمیائی مساوات کی مدد سے پیروکسی ایسیٹائل نائٹریٹس (PANs) کی تشکیل میں نائٹروجن آکسائیڈز اور غیر سوختہ ہائیڈروکاربنز کے کردار کی وضاحت کریں۔	1+1 +1	OR	Give two commercial uses of ethyne. ایتھائین کے کمرشل فائدے بیان کریں۔	1.5 + 1.5
(x)	Explain fat soluble vitamins. چکنائی میں حل پذیر وٹامنز کی وضاحت کریں۔	03	OR	Enlist any three applications of alcohol as Fuel? بطور ایندھن الکوحل کے کوئی تین استعمال درج کریں۔	1+1 +1
(xi)	Outline the importance of polymers in the textile industry (any three). ٹیکسٹائل کی صنعت میں پولیمرز کی اہمیت بیان کریں۔ (کوئی تین)	1+1 +1	OR	Give reactions of ethanoic acid with sodium, sodium carbonate and sodium hydroxide. ایتھانوئک ایسڈ کے درج شدہ کے ساتھ تعاملات کی مساوات لکھیں: الف۔ سوڈیم ب۔ سوڈیم کاربونیٹ ج۔ سوڈیم ہائیڈروآکسائیڈ	1+1 +1

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SECTION – C (Marks 20)

Note: Attempt all questions. تمام سوالات حل کریں۔

Q.3	Describe the following physical parameters that affect the rate of reaction I) Change in mass II) Temperature iii) Formation of gas. درج طبعی عوامل کی وضاحت کریں جو تعامل کی رفتار کو متاثر کرتے ہیں: i کمیت میں تبدیلی ii درجہ حرارت iii گیس کی تشکیل	2+2 +2	OR	Describe industrial applications of ester and carboxylic acid. (Any six) کاربوکسیلک ایسڈ اور ایسٹر کے صنعتی استعمالات کی تفصیل تحریر کریں۔ (کوئی سے چھ)	06
Q.4	$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$ Considering the above chemical equation if 28 g of carbon monoxide (CO) reacts with an excess of oxygen gas, calculate the number of moles of carbon dioxide (CO_2) produced. Also calculate the volume of carbon dioxide (CO_2) produced at room temperature and pressure (RTP), where 1 mole of gas = 24 dm^3. مندرجہ بالا کیمیائی مساوات کو مدِ نظر رکھتے ہوئے، اگر 28 g کاربن مونو آکسائیڈ (CO) آکسیجن گیس کی زائد مقدار کے ساتھ تعامل کرے تو پیدا ہونے والی کاربن ڈائی آکسائیڈ (CO_2) کے مولز کی تعداد معلوم کریں۔ نیز، کمرے کے درجہ حرارت اور دباؤ (RTP) پر پیدا ہونے والی کاربن ڈائی آکسائیڈ (CO_2) کا حجم بھی معلوم کریں، جبکہ: 1 mole of gas = 24 dm^3	3+3	OR	Calculate empirical and molecular formula of a compound having molar mass of 70 g/mol. Also provided with 10.0 g of carbon, 3.0 g of hydrogen and 2.0 g of oxygen. ایک مرکب کے امپریکل اور مالیکیولر فارمولے معلوم کریں جس کی مولر کمیت 70 g/mol ہے۔ مرکب میں 10.0 g کاربن، 3.0 g ہائیڈروجن اور 2.0 g آکسیجن موجود ہیں۔	4+2
Q.5	For the electrolysis of dilute copper (II) sulphate using inert electrodes. I) Identify the products formed at anode and cathode by giving chemical equations ii) The direction of ion movement iii) Observation as the reaction progress غیر عامل الیکٹروڈز (Inert Electrodes) استعمال کرتے ہوئے کاپر (II) سلفیٹ کے ہلکے آبی محلول کی برق پاشیدگی کے لیے: i. اینوڈ اور کیتھوڈ پر بننے والی مصنوعات کی کیمیائی مساوات سمیت نشاندہی کریں۔ ii. آئنوں کی حرکت کی سمت بیان کریں۔ iii. تعامل کے آگے بڑھنے کے ساتھ ہونے والے مشاہدات بیان کریں۔	2+1 +1	OR	Recognize the type of polymerization with complete chemical equation for the formation of polyester molecule by using dicarboxylic acid and a diamine. Also identify the structure and functional group of the repeating unit in the resulting polymer. ڈائی کاربوکسلک ایسڈ اور ڈائی امین استعمال کر کے پولی ایسٹر کے مالیکیول کی تشکیل کے لیے پولیمرائزیشن کی قسم کی مکمل کیمیائی مساوات سمیت نشاندہی کریں۔ نیز بننے والے پولیمر کی ساخت اور فنکشنل گروپ کی (Repeating Unit) میں دہرائی جانے والی اکائی کی نشاندہی کریں۔	2+2
Q.6	Identify the functional group in the following compounds: a. $H_3C-\overset{\overset{O}{\|}}{C}-O-H$ b. $CH_3-C \equiv C-CH_3$ c. $H_3C-\overset{\overset{O}{\|}}{C}-O-CH_3$ d. CH_3CH_2OH درج شدہ مرکبات میں موجود فنکشنل گروپس کی نشاندہی کریں:	1x4	OR	Provide balanced chemical equations of metal with: i. Dilute acid ii. Cold water iii. Steam iv. Oxygen. دھاتوں کی عمومی کیمیائی خصوصیات کو ان کی تعامل پذیری کے حوالے سے درج ذیل کے ساتھ متوازن کیمیائی مساوات کی مدد سے بیان کریں: i ہلکے تیزاب ii ٹھنڈا پانی iii بھاپ iv آکسیجن	1x4

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SUPPLEMENTARY TABLE —— 2SA-II 26005 (New) ——

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40

2nd Annual 2026 — Chemistry SSC-II

Complete solutions • Sections B and C

SECTION B — Q2: Short questions (33 marks)

Q2(i) Increasing reactivity from standard reduction potentials of Mg, Cu and Ni.

Metal	Standard reduction potential
Mg	−2.37 V
Ni	−0.25 V
Cu	+0.34 V

Increasing order: Cu < Ni < Mg

A more negative standard reduction potential generally means the metal loses electrons more readily. Thus Mg is most reactive and Cu least reactive.

OR — Equations: zinc + steam; magnesium + HCl; sodium + cold water.

$Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_2(g)$ [heat]

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

Q2(ii) Interpret heating and cooling curves using kinetic theory.

Heating curve: on a sloping section, average kinetic energy increases; particles move or vibrate faster, and temperature rises. At melting or boiling, temperature is constant: energy increases potential energy by overcoming attractions rather than increasing average kinetic energy.

Cooling curve: on a sloping section, average kinetic energy decreases and temperature falls. During condensation or freezing, temperature stays constant while latent heat is released as stronger attractions form between particles.

OR — Molarity of 98 g H_2SO_4 in 1 dm^3 of water (molar mass 98).

$n = m/M_r = 98/98 = 1.0$ mol

Assuming final solution volume 1 dm^3: $M = n/V = 1.0/1$

$M = 1.0$ mol dm^{-3}

The expected answer assumes 1 dm^3 of final solution. The paper says “1 dm^3 of water”; exact molarity requires the final solution volume.

Q2(iii) $MgCO_3$ contains 15 g Mg. Find CO_2 mass on decomposition.

$MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$ [heat]

Each mole of $MgCO_3$ contains one mole of Mg and releases one mole of CO_2.

$n(Mg) = 15/24 = 0.625$ mol

$n(CO_2) = 0.625$ mol

$M_r(CO_2) = 12 + 2(16) = 44$

$m(CO_2) = 0.625 \times 44 = 27.5$ g

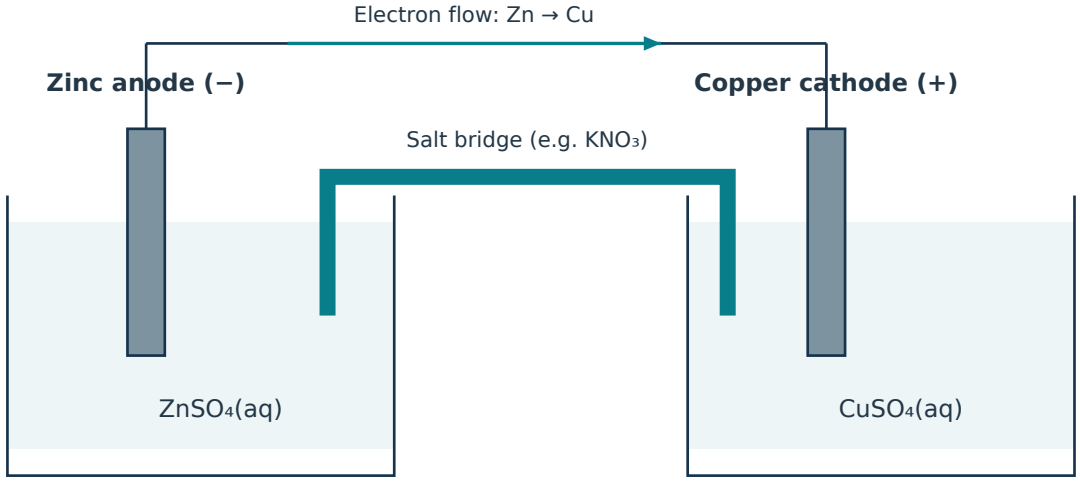
OR — Structural formulae: but-2-ene, propan-1-ol and ethanoic acid.

Compound	Condensed structural formula
But-2-ene	$CH_3–CH=CH–CH_3$
Propan-1-ol	$CH_3–CH_2–CH_2–OH$
Ethanoic acid	$CH_3–C(=O)–OH$

“Prop-1-ol” in the paper should be written as propan-1-ol.

Q2(iv) Daniell cell: anode, cathode and electron direction.

A Daniell cell has a zinc electrode in $ZnSO_4$ solution and a copper electrode in $CuSO_4$ solution. The electrodes are joined by an external wire and the solutions by a salt bridge.



Anions → zinc half-cell; cations → copper half-cell

Anode (negative), oxidation:

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Cathode (positive), reduction:

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Overall:

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

Electrons flow **from zinc to copper** through the external wire. The salt bridge completes the circuit and maintains electrical neutrality: anions migrate towards the zinc half-cell and cations towards the copper half-cell.

Cell notation: $Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$

OR — Structure showing protein formation and peptide linkages.

Amino acids join through **condensation reactions**. The carboxyl group of one reacts with the amino group of another, releasing water and forming a peptide linkage.

$H_2N–CH(R_1)–COOH + H_2N–CH(R_2)–COOH$

$\rightarrow H_2N–CH(R_1)–CO–NH–CH(R_2)–COOH + H_2O$

$$H_2N–CH(R_1)–\boxed{CO–NH}–CH(R_2)–\boxed{CO–NH}–CH(R_3)–COOH$$

Peptide linkages: –CO–NH–

Each linkage forms by condensation, releasing H_2O.

Many amino acids join to form a polypeptide chain, which folds into a protein's structure. The peptide linkage is **–CO–NH–**.

Q2(v) Moles of oxygen in 25 dm³ at room temperature.

Assume room temperature and pressure (RTP), with molar gas volume 24 $dm^3\ mol^{-1}$.

$n = V/V_m = 25/24 \approx 1.04$ mol O_2

OR — Hydrogen–oxygen fuel cell: anode, cathode and overall equations.

For a hydrogen–oxygen fuel cell with an **alkaline electrolyte**:

Anode — oxidation:

$2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$

Cathode — reduction:

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

Overall:

$2H_2 + O_2 \rightarrow 2H_2O$

Electrons flow from the anode to the cathode through the external circuit. Chemical energy is converted into electrical energy.

Q2(vi) Compare hydration and fermentation for making ethanol.

Feature	Hydration	Fermentation
Raw materials	Ethene and steam	Plant-derived glucose
Conditions	≈300°C, 60–70 atm, H_3PO_4 catalyst	≈30–35°C, yeast enzymes, no oxygen
Production	Fast; can run continuously	Slower; usually batch production
Product	Ethanol separated from reaction mixture	Dilute ethanol requiring purification

Hydration: $C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$

Fermentation: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

OR — Three general solubility rules for salts.

1. All **nitrates** are soluble in water.
2. All **sodium, potassium and ammonium salts** are soluble.
3. Most **chlorides** are soluble, except salts such as silver chloride and lead(II) chloride, which are insoluble or sparingly soluble.

Q2(vii) Complete bromoethane + alcoholic KOH and ethene + bromine.

(a) Alcoholic KOH and heat cause elimination (dehydrohalogenation).

$CH_3CH_2Br + KOH \rightarrow CH_2{=}CH_2 + KBr + H_2O$

(b) Bromine adds across the double bond.

$CH_2{=}CH_2 + Br_2 \rightarrow CH_2Br{-}CH_2Br$

The product is **1,2-dibromoethane**.

OR — Justify acidic SO_2 and basic CaO with equations.

SO_2 reacts with a base, forming salt and water; it is **acidic**.

$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

CaO reacts with an acid, forming salt and water; it is **basic**.

$CaO + 2HCl \rightarrow CaCl_2 + H_2O$

Q2(viii) Six fractions of petroleum.

1. Refinery gases
2. Petrol / gasoline
3. Naphtha
4. Kerosene
5. Diesel / gas oil
6. Fuel oil

OR — Distinguish ethene and ethane by chemical test.

Add bromine water at room temperature, away from UV light.

Hydrocarbon	Observation	Explanation
Ethene	Decolourises orange-brown bromine water	Bromine adds across the C=C bond
Ethane	No colour change	Saturated; no addition reaction under these conditions

$CH_2{=}CH_2 + Br_2 \rightarrow CH_2Br{-}CH_2Br$

The product is **1,2-dibromoethane**.

Q2(ix) Formation of PAN: role of nitrogen oxides and unburned hydrocarbons.

Unburned hydrocarbons undergo atmospheric oxidation and can form aldehydes such as ethanal. Sunlight drives the chemistry that generates the radicals involved. The following three reactions lead to PAN:

$$CH_3CHO + OH\bullet \rightarrow CH_3CO\bullet + H_2O$$

$$CH_3CO\bullet + O_2 \rightarrow CH_3C(O)OO\bullet$$

$$CH_3C(O)OO\bullet + NO_2 \rightleftharpoons CH_3C(O)OONO_2$$

The product is **peroxyacetyl nitrate (PAN)**, a component of photochemical smog. It irritates eyes and damages plants. The dot denotes an unpaired electron in a radical.

OR — Two commercial uses of ethyne.

1. **Welding and cutting metals:** ethyne burns in oxygen to produce the very hot oxyacetylene flame.
2. **Chemical manufacture:** raw material for compounds such as chloroethene, used to make PVC.

$$HC{\equiv}CH + HCl \rightarrow CH_2{=}CHCl$$

Q2(x) Explain fat-soluble vitamins.

Fat-soluble vitamins dissolve in fats and are absorbed with dietary fats. They can be stored in the liver and fatty tissues. They are **A, D, E and K**.

Vitamin	Main function
A	Normal vision and healthy epithelial tissues
D	Calcium absorption; strong bones and teeth
E	Antioxidant protection of cells
K	Normal blood clotting

OR — Three applications of alcohols as fuels.

1. **Motor fuel:** ethanol is blended with petrol to make fuels such as gasohol.
2. **Spirit burners and camping stoves:** alcohols provide heat in laboratory and portable cooking equipment.
3. **Fuel cells:** methanol is used in direct methanol fuel cells to generate electricity.

Q2(xi) Three roles of polymers in the textile industry.

1. **Fibre manufacture:** polyester, nylon and acrylic are used for clothes, fabrics and carpets.
2. **Strength and durability:** synthetic fibres resist wear and improve blended-fabric strength.
3. **Useful properties:** polymer fibres can produce lightweight, crease-resistant, elastic or quick-drying textiles.

OR — Ethanoic acid with sodium, sodium carbonate and sodium hydroxide.

(a) Sodium: sodium ethanoate and hydrogen form.

$$2CH_3COOH + 2Na \rightarrow 2CH_3COONa + H_2$$

(b) Sodium carbonate: sodium ethanoate, CO_2 and water form.

$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$

(c) Sodium hydroxide: sodium ethanoate and water form.

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

SECTION C — Long questions (20 marks)

Q3 Change in mass, temperature and gas formation in reaction kinetics.

Temperature directly affects rate. Mass change and gas formation are generally ways of measuring rate, rather than independent factors causing it to change.

(i) Change in mass: if a reaction releases gas that escapes, measured mass decreases. Record mass at regular intervals.

$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

Average rate = mass lost / time taken

A greater mass loss per second means a faster reaction.

(ii) Temperature: increasing temperature raises average particle kinetic energy. Collisions occur more frequently, and a larger proportion have energy at least equal to activation energy. Thus there are more successful collisions per second.

(iii) Formation of gas: collect gas in a gas syringe and measure its volume at regular intervals.

Average rate = gas volume formed / time taken

More gas per second means a faster reaction. When volume stops increasing, no more gas is being produced.

OR — Six industrial uses of esters and carboxylic acids.

Esters:

1. Food flavourings: many esters have fruity smells.
2. Perfumes and cosmetics: pleasant-smelling esters provide fragrances.
3. Solvents: ethyl ethanoate is used in paints, coatings, inks and adhesives.

Carboxylic acids:

4. Food industry: ethanoic acid in vinegar; benzoic acid as a preservative.
5. Soap manufacture: long-chain fatty acids are converted to sodium or potassium salts.
6. Polymer manufacture: dicarboxylic acids are used for polyesters and polyamides, including PET and nylon.

Q4 28 g CO + excess O_2: moles and RTP volume of CO_2.

$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$

(a) Moles:

$M_r(CO) = 12 + 16 = 28\ g\ mol^{-1}$

$n(CO) = 28/28 = 1.0\ mol$

The ratio $CO:CO_2$ is 2:2 = 1:1.

$n(CO_2) = 1.0$ mol

(b) Volume at RTP:

$V = nV_m = 1.0 \times 24 = 24$ dm³

OR — Empirical/molecular formula: 10.0 g C, 3.0 g H, 2.0 g O; molar mass 70.

The printed data are inconsistent. A valid molecular formula cannot be obtained.

Element	Moles	Divide by 0.125
C	10.0/12 = 0.8333	6.6667
H	3.0/1 = 3.000	24
O	2.0/16 = 0.125	1

Multiply by 3 to obtain the formal whole-number ratio:

C:H:O = 20:72:3

Formal ratio: **$C_{20}H_{72}O_3$**.

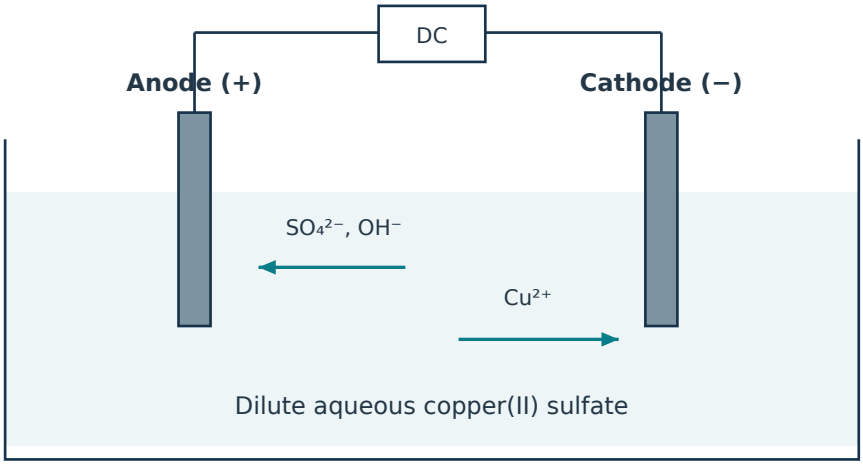
Formula mass = 20(12) + 72(1) + 3(16) = 360

Molecular multiplier = 70/360 ≈ 0.1944

The multiplier must be a positive whole number. The formal ratio also has too much hydrogen for an ordinary neutral C/H/O compound. **The data must be corrected; do not force a formula by incorrect rounding.**

Q5 Electrolysis of dilute $CuSO_4$ using inert electrodes.

Use inert electrodes, such as platinum, immersed in dilute aqueous copper(II) sulfate and connected to a DC supply.



Both electrodes immersed; electrodes do not touch

Cathode (negative) — reduction:

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Copper forms a **reddish-brown coating**, increasing the cathode's mass.

Anode (positive) — oxidation:

$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$

Equivalent form:

$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

Bubbles of **oxygen** form at the anode. Oxygen relights a glowing splint.

Ion movement: Cu^{2+} and other cations migrate towards the cathode; SO_4^{2-} and OH^- migrate towards the anode. Sulfate ions remain in solution rather than being discharged under these conditions.

Overall observations: the blue colour becomes paler as Cu^{2+} ions are removed; the solution becomes more acidic; copper forms at the cathode and oxygen at the anode.

$2CuSO_4 + 2H_2O \rightarrow 2Cu + O_2 + 2H_2SO_4$

OR — Polymer from dicarboxylic acid and diamine, called “polyester” in the paper.

A dicarboxylic acid and a diamine form a POLYAMIDE, not a polyester. Both the stated-monomer answer and likely intended polyester answer follow.

A. Using the monomers actually stated:

A **dicarboxylic acid** (HOOC–R–COOH) and a **diamine** (H_2N–R′–NH_2) form a **polyamide by condensation polymerisation**. A carboxyl group reacts with an amino group, releasing water. Both monomers have two reactive groups, so the chain can continue growing.

–COOH + H_2N– → –CO–NH– + H_2O

General equation, including chain end groups:

n H_2N–R′–NH_2 + n HOOC–R–COOH

→ $H[–NH–R′–NH–CO–R–CO–]_nOH + (2n − 1)H_2O$

$$[–NH–R′–NH–C(=O)–R–C(=O)–]_n$$

Amide linkage: –CO–NH–

R and R′ represent the carbon-containing parts of the monomers.

Functional group: amide linkage, **–CO–NH–**.

Example: nylon-6,6, formed from hexane-1,6-diamine (H_2N–$(CH_2)_6$–NH_2) and hexanedioic acid (HOOC–$(CH_2)_4$–COOH).

Nylon-6,6 repeating unit: $[–NH–(CH_2)_6–NH–CO–(CH_2)_4–CO–]_n$

Applications: strong fibres for ropes, clothing and carpets.

B. If polyester was intended: use a **dicarboxylic acid and a diol**. This is also condensation polymerisation.

–COOH + HO– → –COO– + H_2O

General equation, including chain end groups:

n HOOC–R–COOH + n HO–R′–OH

→ $H[–O–R′–O–CO–R–CO–]_nOH + (2n − 1)H_2O$

$$[–O–R′–O–C(=O)–R–C(=O)–]_n$$

Ester linkage: –C(=O)–O–

R and R′ represent the carbon-containing parts of the monomers.

Functional group: ester linkage, **–C(=O)–O–**.

Example: PET, from ethane-1,2-diol and benzene-1,4-dicarboxylic acid.

Q6 Identify the four functional groups.

Part / compound	Functional group	Class and name
(a) $CH_3–C(=O)–OH$	Carboxyl group, –COOH	Carboxylic acid: ethanoic acid
(b) $CH_3–C≡C–CH_3$	Carbon–carbon triple bond, C≡C	Alkyne: but-2-yne
(c) $CH_3–C(=O)–O–CH_3$	Ester linkage, –COO–	Ester: methyl ethanoate
(d) CH_3CH_2OH	Hydroxyl group, –OH	Alcohol: ethanol

OR — Balanced equations of metals with dilute acid, cold water, steam and oxygen.

(i) Dilute acid: a suitable metal above hydrogen reacts with dilute HCl to form a salt and hydrogen.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Magnesium dissolves with bubbles of hydrogen.

(ii) Cold water: very reactive metals form a hydroxide and hydrogen.

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$

Sodium moves rapidly on the water surface.

(iii) Steam: suitable heated metals form an oxide and hydrogen.

$$Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_2(g) \text{ [heat]}$$

(iv) Oxygen: many metals form metal oxides.

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

Magnesium burns with a dazzling white flame and forms white MgO.

Scientific reference notes

The supplied original papers are reproduced within each section. The worked answers are prepared for Class 10 study, with extra clarification only where it prevents a scientific error.

PAN chemistry: University of York, Master Chemical Mechanism (PAN species and precursor reaction); US EPA, Sources and Control of Volatile Organic Air Pollutants, Student Manual (2002).

https://mcm.york.ac.uk/MCM/species/PAN

https://airknowledge.gov/ILT/TOXC331/Current/CI/03TOXC331_StudentManual_2002.pdf

Chromium plating: US EPA AP-42, Electroplating, Section 12.20. Conventional chromium plating uses insoluble lead-alloy anodes; school-level soluble-anode equations are labelled as a simplified model.

https://www.epa.gov/sites/default/files/2020-11/documents/b12s20.pdf

Fruit ripening: Government of Punjab information on carbide-based ripening distinguishes the textbook acetylene reaction from appropriate commercial use of ethylene.

https://punjab.gov.pk/node/2658

Preparation reminder: learn the chemistry and the method, not incorrect wording or inconsistent data in a printed question.